

MISALLOCATION AND PRODUCT CHOICE

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Abstract

We study the costs of misallocation of inputs between multi-product firms that endogenously choose among heterogeneous products. Misallocation of inputs between firms has been shown to be a significant drag on aggregate productivity: it is especially severe between farms in the agricultural sectors of low-income economies. Existing estimates of its costs have relied on models of single-product firms using a single aggregate production function. Using rich farm-crop-level data from India, we estimate product-level production functions and find that they are meaningfully different from one another and from the aggregate one. We build a general equilibrium model of firm-level misallocation in which multi-product firms (or farms) are able to choose the set and mix of heterogeneous products. Equalizing cross-farm input-market frictions raises aggregate productivity by 137% in the multi-product model and 167% in a conventional single-product model: the single-product model overstates the aggregate productivity cost of misallocation between Indian farms by 22%.

Keywords: misallocation, multi-product firms, agriculture, productivity, distortions

JEL classification: O4, O11, O13, Q1

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1 INTRODUCTION

Misallocation of production inputs between firms is an important driver of cross-country income differences (Hsieh and Klenow 2009; Restuccia and Rogerson 2017). Various market frictions and distortionary government policies prevent inputs from being allocated to their most productive uses: some productive firms would prefer to hire more land, labor, or capital, while some less productive ones would be happy to supply these inputs. However, both parties find themselves unable to do so due to imperfect markets or institutional obstacles, limiting the aggregate productivity of the entire economy. Misallocation is especially severe between farms within the agricultural sectors of low-income countries: allowing farmers to efficiently trade land could triple the aggregate agricultural productivity in the most distorted markets (Chen, Restuccia, and Santaaulalia-Llopis 2022). Some of the estimated aggregate cost of misallocation can be explained by mismeasurement and model misspecification (Bils, Klenow, and Ruane 2021; Gollin and Udry 2021; Asker, Collard-Wexler, and De Loecker 2014). At the same time, a large portion of this cost reflects real maligned frictions: poor land property rights enforcement, inefficient land distribution institutions, distortionary government policies, poor market access, and more (Gottlieb and Grobovšek 2019; Chen 2017; Krishnaswamy 2018; Le 2020; Morando 2023). Understanding both the costs and the causes of misallocation is crucial for understanding the obstacles to aggregate development, especially in agriculture, where cross-country productivity differences are particularly stark (Gollin, Lagakos, and Waugh 2014).

The misallocation literature has focused on studying single-product firms. The literature on agricultural misallocation has further modeled all farms as producing the same homogeneous agricultural good. Farms in the data, however, have wildly different selections of crops: we find that even the most commonly grown crop in India is produced by only one-quarter of all farmers. Moreover, two-thirds of Indian farmers grow two or more different products within a year.

In this paper, we study the aggregate costs of misallocation of inputs between multi-product farms that choose among heterogeneous products. When products have different production functions, they are exposed differentially to misallocative market distortions. We quantify how these differences affect the cost of misallocation by comparing a multi-product and a single-product model fitted to the same farm-level data.

We explore these mechanisms in rich farm-crop-level data from India. We document significantly heterogeneous product choice across farmers, across seasons, and frequent multi-product behavior by individual farmers. Next, we estimate crop-specific production functions using instruments derived from shocks to other plots within a farm to address the simultaneity bias. We find that many crops have significantly different input elasticities from one another—and from the traditional aggregate agricultural good production function—suggesting that products in the data are meaningfully heterogeneous for the purpose of quantifying misallocation. A placebo exercise inspired by randomization inference shows that the observed heterogeneity is exceedingly unlikely to be a lucky draw generated by multiple testing.

We build a general equilibrium model of heterogeneous firms facing misallocative distortions. The main novelty of the model is that firms (or farms) choose among multiple products (or crops) with heterogeneous production functions and can elect to produce any number of products at once. The model provides a mapping from the observable farm-crop-level input and output choices to the unobservable fundamental distortions that are the model’s stand-in for all kinds of institutional and market frictions. Once these distortions are recovered from the data, the model provides a quantification of the aggregate cost of misallocation induced by the frictions.

We decompose the difference in reallocation gains through a sequence of auxiliary models that progressively remove five features of the multi-product economy. First, we equalize returns to scale across crops while preserving their relative factor intensities. Second, we pool factor intensities as well. Third, we freeze each farm’s allocation of inputs across its existing crops. Fourth, we replace crop-specific demand and relative prices with a homogeneous consumption good. Finally, we aggregate farm-crop observations to farm level and recover productivity and distortions using the single-product model. Each model reproduces the same current output and applies the same reallocation policy rule to its own recovered fundamentals.

In our benchmark exercise, equalizing cross-farm input-market frictions raises agricultural productivity by 137% in the multi-product model and 167% in the single-product model. The single-product model therefore *overstates* the gain by 30 percentage points, or 22% relative to the multi-product estimate. Demand aggregation contributes 24.7 percentage points to the gap, and producer aggregation contributes another 4.3 points. Together, these two steps account for 96% of the gap between models. A partial reallocation that brings distortions across India down to the levels of the least-distorted state raises productivity by 34% in the multi-product model and

43% in the single-product model, a relative overstatement of 27%.

The single-product procedure also identifies apparent misallocation in an economy without cross-farm input-market frictions. Taking the multi-product benchmark counterfactual as new data and repeating the single-product measurement and reallocation procedure yields a further predicted gain of 12.3% of that counterfactual's output. The predicted gain remains 11.5% when the source allocation also equalizes within-farm wedges. These exercises isolate a measurement problem: a single-product model applied to a multi-product world picks up misallocative frictions where there are none.

Estimating the cost of agricultural misallocation in farm-level data is the subject of a growing literature. The foundational result on the severity of misallocation in the agriculture of low-income countries was established by [Chen, Restuccia, and Santaaulalia-Llopis \(2022\)](#). Some papers focus on identifying a particular institutional source of distortions in the data and then using a model to quantify the aggregate cost of those distortions ([Chen 2017](#); [Gottlieb and Grobovšek 2019](#); [Adamopoulos et al. 2022](#)). Others seek to capture all sorts of market frictions and government distortions generally, and make methodological contributions to estimating their cost ([Gollin and Udry 2021](#); [Aragón, Restuccia, and Rud 2022](#); [Ayerst, Brandt, and Restuccia 2023](#)). Our paper contributes to the literature by departing from its assumption of a single homogeneous agricultural product. We estimate crop-specific production functions and build a model of multi-product farms endogenously choosing among those crops. We thus contribute both to the question of measuring distortions in the data and to the question of quantifying the aggregate cost of these distortions.

The effects of government policy or technological changes on country-level crop allocation have been explored before. [Blanco and Raurich \(2022\)](#) show that falling capital costs cause the aggregate allocation to shift toward capital-intensive crops. [Le \(2020\)](#) finds that land use policies in Vietnam generate aggregate misallocation by favoring rice over other crops. [Krishnaswamy \(2018\)](#) documents that Indian price support for staple crops distorts the market in favor of rice and wheat. We contribute to this literature by exploring the interplay between distortions and product choice not at the aggregate level, but at the level of individual farmers. The model we devise then allows us to map the micro-level distortions and farm responses to aggregate productivity: this mapping would apply to any individual friction the literature has studied before. A related contribution is by [Ayerst, Brandt, and Restuccia \(2023\)](#), who incorporate crop choice into their model of farm-level misallocation with farm

dynamics. However, they model single-product farms producing the same homogeneous agricultural good: crops are heterogeneous only in their productivities and the productivity-maximizing choice is prevented only by direct land use restrictions. Our model instead incorporates multi-product farms choosing between crops with heterogeneous production functions, opening up several channels of interaction between product choice, estimated distortions, and resulting misallocation.

Multi-product firms have been discussed in the manufacturing misallocation literature by [Jaef \(2018\)](#) and [Wang and Yang \(2023\)](#). In both papers, firms face distortions and can choose the number of products to manufacture among varieties that have the same production function but different productivities. Both papers then calibrate their models to match cross-sectional distribution moments in the data. First, we contribute to this literature by exploring product choice between products with different productivities and different input elasticities, allowing technology heterogeneity to affect both inferred distortions and responses to reallocation. Second, we contribute by extending [Hsieh and Klenow \(2009\)](#)'s framework of mapping firm-level observables to model-implied firm-level fundamentals to the case of multi-product firms. This allows us to reproduce each observed firm (or farm) with its model analog exactly, without having to calibrate distributions of firms to aggregate moments.

Our contribution on the interaction between misallocation and endogenous product choice is not conceptually restricted to agriculture. However, there are several reasons why agriculture is the perfect setting to begin studying this interaction. First, firm-product-level (or farm-crop-level) inputs and outputs are far more feasible to measure in agriculture than in other sectors. Farmers are better able to estimate the amount of land, labor, and intermediate inputs applied to each crop. Such data is also more commonly collected: we use India's REDS, but multiple other farm-level surveys gather crop-level information. Second, products (crops) are far more homogeneous in agriculture than in other sectors. Outside of very specific sub-sectors, varieties produced by different manufacturers operating in the same sub-sector can be vastly different. In agriculture, however, specimens of the same crop grown on different farms can be reasonably approximated as homogeneous. This facilitates the task of estimating product-level production functions and exploring systematic differences in input intensities between products. For these reasons, we choose to focus our initial exploration of endogenous product choice on agriculture. But the mechanism applies more broadly, and insights gleaned from the agricultural setting of a developing country will apply also to the manufacturing sectors of more developed

countries.

The paper proceeds as follows. Section 2 discusses the farm-level data we use and the institutional context of Indian agriculture. Section 3 documents the heterogeneous product choice among Indian farmers. Section 4 estimates crop-level production functions. Section 5 describes the multi-product firm model of misallocation. Section 6 quantifies misallocation and compares it across models. Section 7 concludes.

2 BACKGROUND AND DATA

2.1 BACKGROUND

Indian agriculture is very diverse and one of the largest in the world in terms of arable lands.¹ Around half of Indian agriculture is rainfed, although the share of the irrigated area has improved over the years. The agricultural sector employs 58 percent of households and is a major source of income and livelihood for the majority of rural households (Gulati and Juneja 2022). It is worth noting that women make up 30 percent of total cultivators and 41 percent of agriculture laborers. However, women only manage and operate 14 percent of the total operational holding (Chanana-Nag and Aggarwal 2020).

LAND MARKET AND TENANCY STRUCTURE. The Indian agricultural system was a zamindari system during the colonial era, wherein lands were in the hands of the few who had the right to lease them out and collect revenue (Banerjee and Iyer 2005). However, after 1947, the government of India brought several land and tenancy-related reforms that changed the structure of ownership and leasing rights. These reforms varied in scale, timing, and implementation across Indian states (Deininger, Jin, and Nagarajan 2008).

The majority of agricultural land in India is privately owned by households. Small-holder farming is widespread. In 1999-2000, the average farm size was 5.3 acres per household, which declined to 4.4 acres in 2007-08 and further to 2.6 acres in 2015-16. This decrease in land holding size can be attributed to family splintering (Andrew D Foster and Mark R Rosenzweig 2002). Land is seldom sold and bought: more than 94 percent of land owners inherited their land. Furthermore, only 12 percent of agri-

1. India possesses more than 150 million hectares of arable land and comes second after the USA.

cultural cultivators participate in the land rental market by renting land either in or out.² Low participation in the land market hinders productivity as it limits the reallocation of land to productive farmers (Deininger et al. 2003; Bolhuis, Rachapalli, and Restuccia 2021).

AGRICULTURAL SEASONS. There are three major agricultural seasons in India, namely Kharif (monsoon), Rabi (spring/winter), and Zaid (summer/dry). The Kharif season typically spans from May/June to early October, during which crops like paddy (rice), maize, sugarcane, jute, cotton, groundnut, and turmeric are cultivated. The Rabi season takes place during winter, with crops being sown between October and December and harvested from March to April. Major crops of the Rabi season are wheat, barley, peas, gram, and mustard. Finally, the Zaid season, the shortest of the three, occurs from March/April to May/June; fruits and vegetables are grown during this season.

2.2 DATA

Our primary data source is the Rural Economics and Demographic Survey (REDS), which was conducted by India's National Council of Applied Economic Research in 1971, 1981, 1999, and 2007-08.³ The first round was collected in 1971. On each subsequent round, all households (or their descendants) living in the same village were surveyed again, along with a random selection of new replacement households (Vashishtha 1989).⁴ The survey is nationally representative of rural India.

For this analysis, we use REDS 2007-08 (hence REDS07), covering a sample of 8,659 households across 242 villages in 17 major states in 2007/08. REDS07 collects detailed information on agricultural inputs and outputs at the level of plot-season-crops. We restrict our attention to this round because it is the only one in which agricultural data was collected at the plot level, which is necessary for our production function identification strategy. Household and village-level modules of the survey were administered during 2007-08. Of the surveyed households, 4,803 cultivated land: we treat each such household as a farm and restrict our sample to them. In total, these farmers cultivated 10,318 plots.

2. Source: Rural Economic and Demographic Survey (REDS) 2007-08.

3. Prior to 1981-82, the survey was known as the Additional Rural Income Survey (ARIS).

4. In 1971, the sampling frame included one district per state in the Intensive Agricultural District Program (IADP) and a random sample of other districts.

3 HETEROGENEOUS PRODUCT CHOICE BY INDIAN FARMS

CROP CHOICE IS HETEROGENEOUS. Indian farmers vary greatly in which crops they choose to produce. Figure 1 shows the 10 most commonly produced crops in India. Rice is the most popular crop—yet it is grown only by 26% of farmers. Wheat is a close second, but all other crops are far behind: Indian farmers have little overlap in crops they choose to grow.⁵

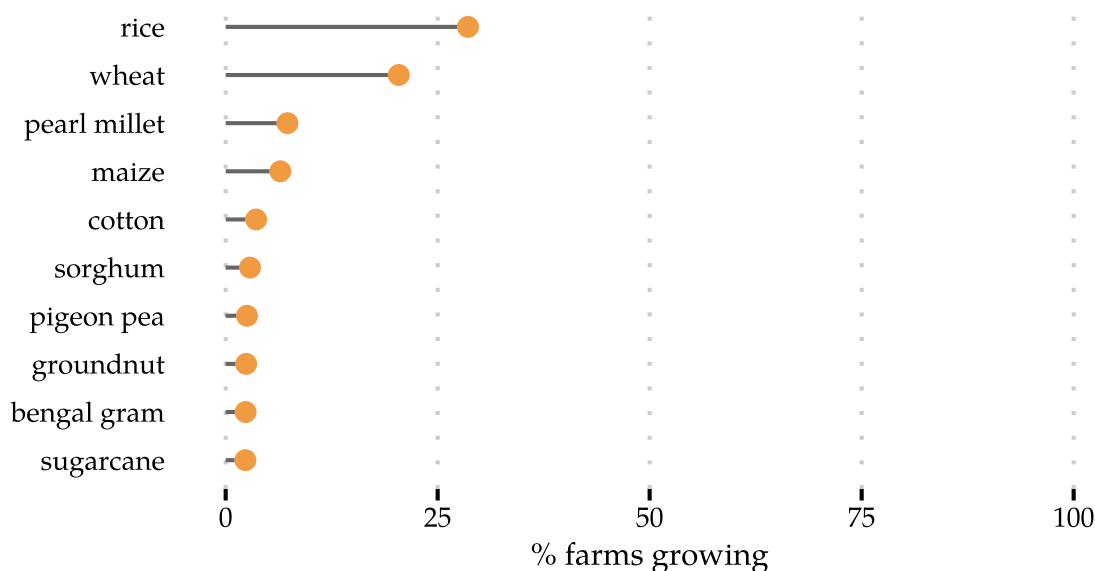


FIGURE 1: Top 10 crops by % of farm-seasons growing them

MANY FARMS GROW MULTIPLE PRODUCTS. Figure 2 displays the share of output produced by farms that grow different numbers of crops. The top bar counts crops that were grown by each farm within the survey year, even if the crops were grown in different seasons: over 80% of output is produced by farmers that grow multiple crops within a year. The bottom bar of Figure 2 treats farm-season combinations as separate entities, thus only counting crops that were all grown in the same season: two-thirds of output is produced by farmers that grow nothing else in the same season, and one-third is produced by multi-crop farmers. Farmers are more specialized within each season than across, although the extent of multi-crop behavior is significant in both cases.

5. Appendix Figure A.1 shows that the ranking of crops by their share in aggregate output has a similar shape, although the order of crops does change.

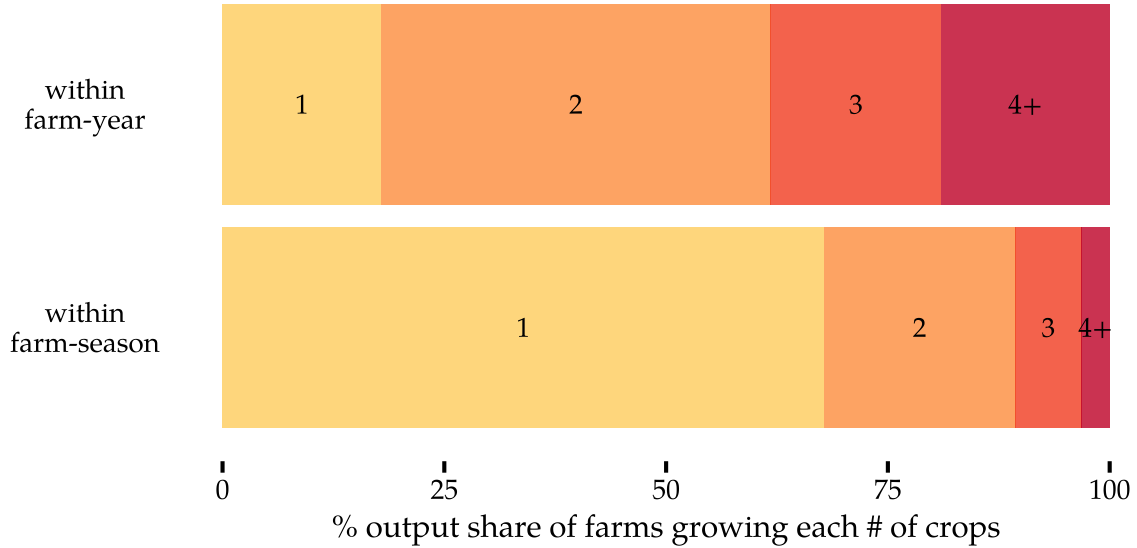


FIGURE 2: Share of output produced by farms growing each number of crops (within a year or within each season alone)

These results have two potential implications for modeling and quantifying misallocation. First, they suggest that production should be split by season when quantifying misallocation, since crops grown in different seasons may be meaningfully different and would affect misallocation estimates if this heterogeneity is not taken into account. Second, the magnitude of multi-crop production is non-negligible both within and across seasons, suggesting that not only modeling heterogeneous crop choice but also the simultaneous cultivation of multiple crops may be important.

At the same time, the apparent heterogeneity of product choice across Indian farmers, across seasons, and the presence of multi-cropping may be of little consequence for estimating misallocation if different crops have similar production functions and would thus interact with distorted input markets the same way. In the next section, we estimate crop-specific production functions to test whether this is the case.

4 ESTIMATING PRODUCT-SPECIFIC PRODUCTION FUNCTIONS

The preceding section documented that Indian farmers engage in heterogeneous multi-product behavior. In this section, we investigate whether these products are meaningfully different in their production functions.

4.1 SPECIFICATION

To estimate input elasticities, we estimate the following crop-specific Cobb-Douglas production function:

$$y_{f,i,t} = z_{f,i,t} l_{f,i,t}^{\gamma_i} x_{labor,f,i,t}^{\alpha_{labor,i}} x_{inter,f,i,t}^{\alpha_{inter,i}}$$

$y_{f,i,t}$ is the gross physical output of farm f for crop i in season t , weighted with market prices (this weighting is irrelevant at the estimation stage).⁶ l is the land input with crop-specific elasticity γ_i . The land input is adjusted for quality, as described in Section 4.3 below. x_{labor} is the labor input with elasticity $\alpha_{labor,i}$, measured in the number of days worked, including both hired and family labor. x_{inter} is the intermediate input with elasticity $\alpha_{inter,i}$, measured as total expenditure on seeds, fertilizer, irrigation, rented machinery, and draft animals. $z_{f,i,t}$ is the farm-season-crop-specific productivity measure that is not directly observed. After taking logs, we estimate the following regression specification:

$$\log y_{j,i,t} = \gamma_i \log l_{j,i,t}^{\gamma_i} + \alpha_{labor,i} \log x_{labor,j,i,t} + \alpha_{inter,i} \log x_{inter,j,i,t} + \epsilon_{j,i,t} \quad (1)$$

for each crop i on plot j , which is the smallest unit of variation in the data. In our data, we observe more than 50 crops grown all across India. For ease of estimation, we group them into five categories: rice; wheat; other cereals; pulses; vegetables, fruits, and oil seeds.⁷ In all that follows, crop-specific production functions refer to these five merged crops.

4.2 IDENTIFICATION

Estimating the production function using ordinary least squares (OLS) as specified in Equation 1 is not consistent since plot-specific unobserved productivity is likely to be correlated with observed input choices. Hence, our estimation strategy follows the idea developed in [Gollin and Udry \(2021\)](#). We estimate Equation 1 using the two-stage least squares method (2SLS), where instruments are developed following the shadow price approach in [Singh, Squire, and Strauss \(1986\)](#) and [Gollin and Udry \(2021\)](#). The idea is simple: within a farm, shocks to plots $k \neq j$ change the shadow

6. Although value-added production functions are common in the misallocation literature, they have a poor theoretical basis and can significantly overstate the productivity dispersion ([Gandhi, Navarro, and Rivers 2017](#)). For this reason, we elect to work with a gross output production function.

7. See Appendix Table A.3 for the composition of these groups.

price of inputs on plot j , providing variation in plot-level input prices that can be used as instrumental variables for input allocations on plot j . The identification of the production function rests on two assumptions: first, that observed shocks to farm f 's plot k affect the shadow price of inputs on f 's plot j ; second, that observed shocks affecting input demand on plot k are not correlated with unobserved shocks affecting input demand on plot j , conditional on observed shocks to j . The survey collects information on a wide array of agricultural, health, and social shocks at the household level. [Gollin and Udry \(2021\)](#) posit that these shocks will have different effects on individual plots depending on the physical features of each plot. Therefore, the household-level (or farm-level) shocks interacted with observed objective plot-level soil features provide the necessary plot-level variation, which serve as instruments for plot-level input choices. Of the three inputs, land deserves special treatment due to its centrality to agricultural production: we turn to it next.

4.3 LAND INPUT MEASURE

There are two commonly used ways of measuring the quantity of the land input in agricultural survey data similar to India's REDS. The first is to measure it with the physical area of cultivated land. This quantity can be measured very precisely and objectively but lacks any information on the quality of land. The second is to measure it with the reported market price of cultivated land, which should capture quality differences between plots. However, land markets in India (and many other developing countries) are extremely under-developed, and so the market price reported by farmers is likely to be exceptionally noisy and mismeasured: in the absence of a market, farmers' ability (and need) to estimate the market price of their land is likely to be poor.

Instead of following either of these two approaches, we settle on a compromise solution that attempts to combine the best of both. We measure the land input on each plot with its physical area weighted by a quality index: price per acre predicted by a statistical model using only the observed objective characteristics of the plot as predictor variables. The objective characteristics measured in the data and included as predictors are various qualities of soil (its type, color, salinity, depth, percolation, and ease of drainage) and objective measures of irrigation access (presence and features of nearby irrigation wells and canals).

The task of constructing the land quality is effectively a regression. Instead of an

ordinary least squares regression (OLS), however, we employ random forests (RF): a supervised machine learning algorithm developed by [Breiman \(2001\)](#). The random forest model “grows” a large number of decision trees, with each tree using a random bootstrap sample of the data, and aggregates the predictions of individual trees.⁸ Unlike OLS, RF does not require the econometrician to pre-specify the interactions and non-linearities expected in the data: it searches for them on the fly and adapts to what it finds in the data. Furthermore, random forests tend to outperform OLS and many other machine learning techniques in out-of-sample predictive performance, reducing the danger of overfitting the noise in reported market price ([Varian 2014](#); [Mullainathan and Spiess 2017](#)).

To evaluate both methods, we estimate an OLS regression and an RF regression using a training sample that randomly selected two-thirds of our data, with the remaining one-third reserved for the test sample. Both methods regress log price per acre on the observed physical characteristics of the plot. The OLS regression includes a complete set of two-way interactions. The RF’s parameters m_{try} and $nodesize$ were tuned with k -fold cross-validation (with $k = 3$), although tuning provides only a small improvement in predictive performance relative to conventional defaults.⁹

Once the two methods have been estimated, we evaluate their performance on the reserved testing sample: results are presented in Table 1. The random forest has both a lower mean squared error and a higher R^2 than OLS. Due to RF’s better out-of-sample performance, we elect to use it to construct the land quality index: predicted price per acre computed for the whole sample.

Because the random forest captures over half of the observed variation in reported price using nothing but observed objective plot features, it likely captures much of the underlying variation in land quality. Because we used predicted prices and took several steps to minimize overfitting (bootstrapping within RF, k -fold cross-validation, and train/test data split), this captured variation is driven by population-level relationships between plot characteristics and market price, rather than the noise in any

8. See [Biau and Scornet \(2016\)](#), [Schonlau and Zou \(2020\)](#), or [Ziegler and König \(2014\)](#) for excellent reviews.

9. Each tree within a random forest is constructed by repeatedly finding the predictor variable and its cutoff value at which to split the data into two branches that maximize the goodness of fit of the branching. m_{try} sets the number of candidate predictors randomly drawn to be considered for branching at each node when constructing each tree. $nodesize$ sets the cutoff number of observations per node at which further splitting stops. The third free parameter in a random forest is the number of trees: we set it high enough that further increases do not improve predictive performance. See any of the RF reviews cited above for details on their operation and the role of these parameters. See [Bischi et al. \(2021\)](#) for an overview of k -fold cross-validation and other tuning techniques.

TABLE 1: Out-of-sample performance of OLS and RF in predicting land price

	Ordinary Least Squares	Random Forest
Mean Squared Error	0.61	0.49
R^2	0.39	0.51

Note. The table presents two metrics of out-of-sample performance of Ordinary Least Squares (OLS) and Random Forest (RF) regressions in predicting log price per acre of a plot with observed physical characteristics of the plot. Two-thirds of plot-level observations were used for estimation and one-third was reserved as a test sample to compute the presented out-of-sample performance metrics.

individual farmer's measurement.

4.4 PRODUCTION FUNCTIONS ARE HETEROGENEOUS ACROSS PRODUCTS

Complete results from production function estimation are shown in Appendix Table A.1. Figure 3 visually summarizes the estimated production function elasticities, including 95% confidence intervals. The first panel displays the input elasticities when a single production function is estimated on all crops pooled together: the commonly used approach. In this case, the results are in line with the literature's findings: land elasticity γ is the highest at 0.43, intermediates elasticity α_{inter} is slightly lower at 0.35, labor elasticity α_{labor} is 0.19. The sum of the three elasticities is 0.97, insignificantly different from 1, indicating that the production function exhibits slightly decreasing or roughly constant returns to scale.

As the following panels indicate, however, crop-specific production functions are heterogeneous across crops. Rice is significantly more land-intensive than the aggregate production function would indicate. Wheat is more intermediates-intensive than rice or the aggregate function, and its returns to scale are increasing. Cereals other than wheat are more in line with the aggregate elasticities but have significantly decreasing returns to scale. For pulses, all three inputs are similarly important. Vegetables, fruits, and oils are more labor-intensive than any other crop, and exhibit decreasing returns to scale.

Pairwise comparisons of a given input's elasticity across crops indicate that many pairs of crops have significantly different elasticities from one another: see Figure 4. For each input elasticity, there are $\binom{5}{2} = 10$ crop pairs that can be formed out of five crops. Of the $3 \times 10 = 30$ combinations across the three input elasticities, 9, 13, and

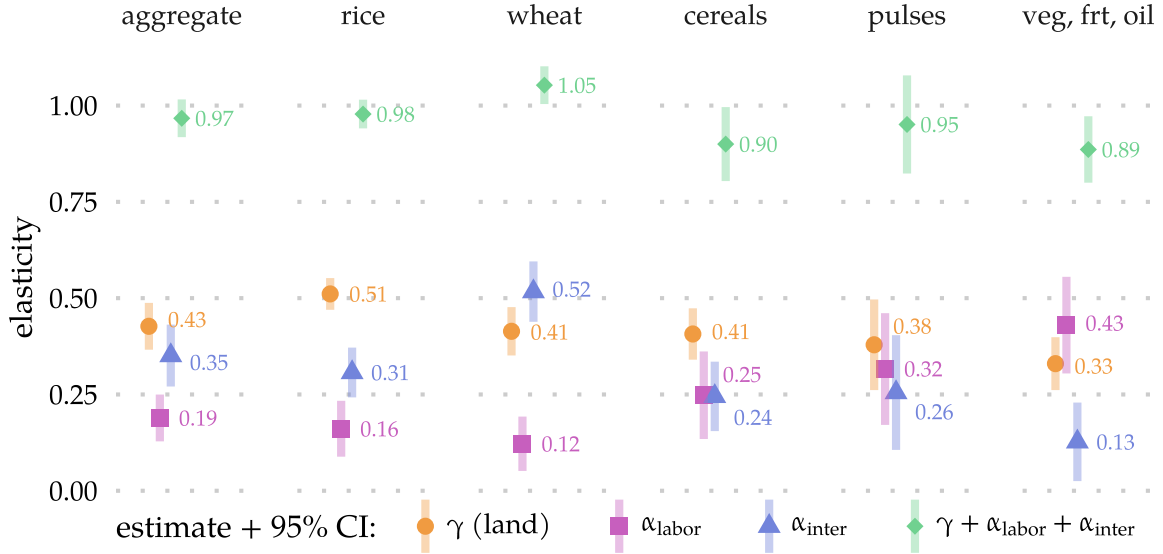


FIGURE 3: Estimated production function input elasticities

Note. The figure visualizes the estimated input elasticities reported in Table A.1. γ is the land, α_{labor} the labor, and α_{inter} the intermediates elasticity. The sum of elasticities $\gamma + \alpha_{labor} + \alpha_{inter}$ reflects the returns to scale.

17 differ significantly at the 1%, 5%, and 10% levels, respectively. Production functions are heterogeneous across products, and a single aggregate production function provides only a poor approximation of any one individual crop.

PLACEBO TEST. Comparing three input elasticities across ten crop pairs entails 30 simultaneous hypothesis tests. Some equality tests may therefore be rejected by chance even if the crop categories contain no systematic production function heterogeneity. To assess whether repeated testing and partitioning the data into thinner slices could produce the observed number of rejections by chance, we conduct a placebo exercise inspired by randomization-based inference.

First, we use the 30 pairwise coefficient equality tests described above as a summary measure of production function heterogeneity. In the actual crop assignment, 17 of these tests reject coefficient equality at the 10% level.

Next, we repeat the full estimation and testing procedure for 1,000 randomly permuted crop assignments. Each permutation preserves the number of observations in every crop category but randomizes the association between observations and crop

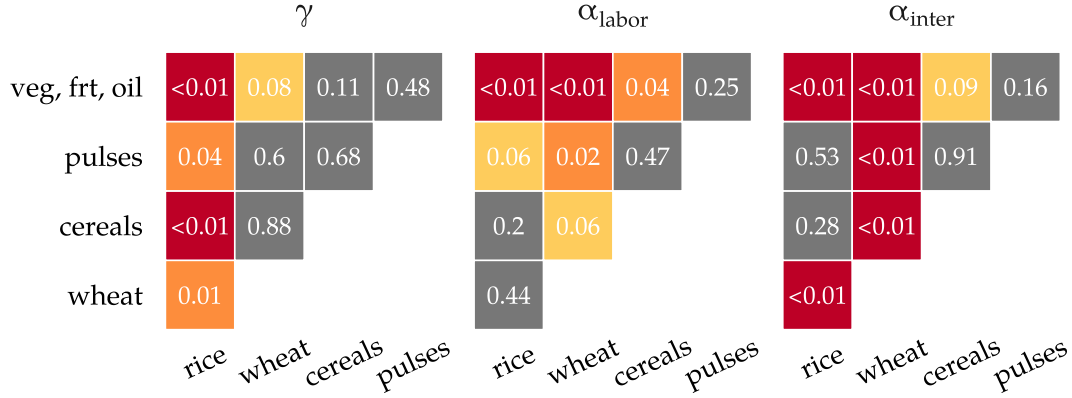


FIGURE 4: P-values of pairwise coefficient comparisons across crop-level production functions

Note. P-values are obtained from tests of equality of input elasticities between each pair of crops in Appendix Table A.1. Color-coded significance levels for rejecting coefficient equality within each crop-pair-input: $p < 0.1$, $p < 0.05$, $p < 0.01$.

categories, creating a “placebo” crop assignment. Re-estimating the five crop-specific production functions and counting the rejected equality tests for every permutation produces the null distribution shown in Figure 5.

Of the 1,000 random permutations, only two match the 17 rejections in the actual crop assignment, and none exceed it. The corresponding right-tail randomization p -value, using the standard plus-one correction, is $3/1001 \approx 0.003$. Random partitions of the data are therefore exceedingly unlikely to produce production functions as heterogeneous as those estimated across actual crops. The observed heterogeneity is not a lucky draw generated by multiple testing.

To quantify the relevance of this heterogeneity for the cost of misallocation, we turn to the model.

5 MODEL

The model pursues three objectives. The first is to provide a framework for thinking about the production decisions of multi-product farmers in the presence of misallocative distortions. The second is to provide a mapping from observable farm-level input and output choices to unobserved distortions these farms must be facing. The

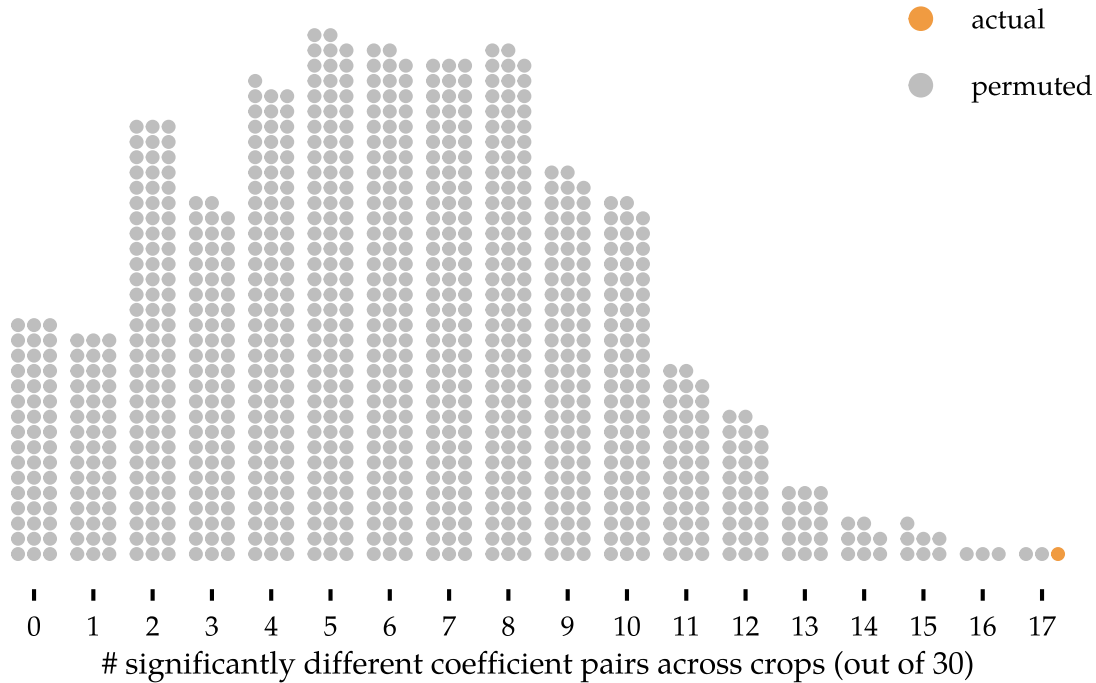


FIGURE 5: Pairwise coefficient equality tests for randomly permuted crop assignments

Note. The figure compares the number of input-elasticity equality tests rejected at the 10% level under the actual crop assignment with the rejection counts from 1,000 size-preserving random permutations of crop assignments. Each assignment produces 30 tests: one for every combination of three inputs and ten crop pairs.

third objective—and the ultimate goal of the paper—is to quantify the aggregate cost of misallocation induced by distortions and to tease out the role of product choice in explaining this cost.

To achieve these objectives, we build a model of farm-level misallocation in which multi-product farmers make production decisions over heterogeneous products. It builds upon the single-product firm model of [Hsieh and Klenow \(2009\)](#).¹⁰ Because their model was applied to data on manufacturing firms, it assumed monopolistic competition and a constant-returns-to-scale production function. Because agricultural markets are populated with small producers growing fairly homogeneous crops, we assume perfect competition instead, in line with the simpler single-product agricultural misallocation framework of [Chen, Restuccia, and Santaaulalia-Llopis \(2022\)](#).

10. Which in turn is closely related to [Restuccia and Rogerson \(2008\)](#).

Instead of assuming constant or decreasing returns to scale for all crops, we rely on our estimated returns to scale for each crop.

5.1 FARM PROBLEM

The economy is populated by F heterogeneous farmers making production decisions over N heterogeneous crops.

Consider the problem of a profit-maximizing farm f . The farm produces quantity $y_{f,i}$ of each crop i and sells it at market price p_i . The farm has idiosyncratic productivity $z_{f,i}$ in each crop. It chooses how much of each input to hire or allocate to each crop.

INPUTS. Inputs $g = 1, \dots, G$ are flexible: the farm can hire each at market rate r_g . The flexible input elasticity $\alpha_{g,i}$ is input-crop-specific. In our calibration, there are two flexible inputs ($G = 2$): labor and intermediates. The farm chooses quantity $x_{f,g,i}$ of each input g to allocate to each crop i .

Land input l is available to the farm in fixed supply L_f : the farm needs to decide how to split this endowment between available crops, assigning $l_{f,i}$ to each. The crop-specific land elasticities are γ_i . There are three reasons for modeling land as a fixed input. The first is that it is a reasonable approximation of the state of the land market in India. Land is rarely purchased, sold, or rented: it is usually simply inherited. The second is that having one input be in fixed supply generates interdependent crop production: a change in parameters of crop i will lead to a change in the shadow cost of the fixed input (land) and thereby impact the input and output decisions the farm makes on crops $j \neq i$.¹¹ The third is that an input in fixed supply justifies the production function identification assumptions we made in Section 4: land l and its shadow cost λ_f provides the link through which observable shocks to one plot can affect the shadow cost of inputs on other plots.

Inputs are combined with Cobb-Douglas production technology to produce each crop i : $y_{f,i} = z_{f,i} l_{f,i}^{\gamma_i} \Pi_g \left(x_{f,g,i}^{\alpha_{g,i}} \right)$.

DISTORTIONS. Misallocative frictions in this class of models are captured by distortion terms $\tau_{f,g}$. These are idiosyncratic to the farm and operate as taxes or subsi-

11. The importance of modeling interdependent crop production with a fixed input rather than any alternative mechanisms is discussed in [Just, Zilberman, and Hochman \(1983\)](#) and [Shumway, Pope, and Nash \(1984\)](#).

dies. If farm f 's $\tau_{f,g}$ is higher for input g than for input h , the farm is facing stronger frictions in input g 's market, and will hire a relatively higher quantity of h instead. If farm f 's frictions $\tau_{f,g}$ for all inputs are higher than those of farm e , then farm f will hire less of every input and will operate at an inefficiently small scale compared to farm e . This simple framework has been used in the misallocation literature to capture all kinds of market frictions and government distortions that firms may be suffering from. Any variation in input ratios or firm scales that is unexplained by the observed production functions and productivities can be replicated by these distortion terms.

Furthermore, once we add multi-product firms into the model, we have to add another layer of distortions. $\tau_{f,g,i}$ and $\tau_{f,l,i}$ are farm-input-crop-specific distortions that can fit unexplained variation in input ratios between crops of a single farm.

Finally, while land input l does not have an explicit distortion term like $\tau_{f,l}$ in the model, land is still de facto distorted since its quantity is fixed. Any variation in the shadow cost of land λ_f between farms reflects implicit τ -like frictions that prevent the farmers from operating an efficient amount of land. As long as land is not distributed in precisely the right way to equalize λ_f between farmers, it is distorted, with the dispersion of λ_f reflecting the severity of distortions.

ADDITIONAL CONCAVITY. When crop-specific production functions have decreasing returns to scale, model farms choose to grow multiple crops at once. However, most of the crop-specific production functions estimated in Section 4.4 exhibit returns to scale that are close to constant, generating little multi-cropping in the model unless frictions are extreme: production concavity alone struggles to explain the extent of multi-product behavior in the data. To allow for a range of other mechanisms that may incentivize farmers to grow multiple crops, we raise the crop-level revenue within the farmer's objective function to some power $\eta \leq 1$. When $\eta < 1$, this generates additional concavity in crop-level revenue. Introducing this parameter allows the model to parsimoniously capture mechanisms such as risk (farmers multi-crop to hedge against uncertain revenue yields in each individual crop), diet diversification by subsistence farmers (subsistence farmers grow multiple crops to satisfy their love of variety), and market power (farmers with some market power in their local market face downward sloping demand for each crop, generating revenue that is concave in inputs). Appendix B shows that the latter two mechanisms, if modeled explicitly, generate a farm's problem that is exactly equivalent to the $\eta < 1$ case. Another reason to allow for $\eta < 1$ is that it takes the burden of generating the observed distribution

of farm-crop revenues away from τ frictions, leading to more conservative misallocation estimates, as Section 5.4 discusses below.

FIXED COST. For each crop i that the farm chooses to produce ($y_{f,i} > 0$), it pays a fixed cost ω . Without this feature, farmers would produce all possible crops, with some of them grown in negligible quantities: introducing the fixed cost makes such marginal crops unprofitable and leads the farm to drop them entirely. We introduce the fixed cost to reproduce the observed distribution of the number of crops grown per farm, discussed in Section 3.

PROBLEM. The complete problem of the farm is:

$$\max_{\{y_{f,i}, l_{f,i}\}_i, \{x_{f,g,i}\}_{g,i}} \sum_{i=1}^N (p_i y_{f,i})^\eta - \sum_{g=1}^G r_g \tau_{f,g} \sum_{i=1}^N \tau_{f,g,i} x_{f,g,i} - \sum_{i=1}^N \omega \cdot 1[y_{f,i} > 0] \quad (2)$$

s.t.

$$y_{f,i} = z_{f,i} l_{f,i}^{\gamma_i} \Pi_g \left(x_{f,g,i}^{\alpha_{g,i}} \right) \quad (3)$$

$$\sum_{i=1}^N l_{f,i} \tau_{f,l,i} = L_f \quad (\lambda_f) \quad (4)$$

The solution and the computational algorithm are discussed in Appendix C.1.

5.2 GENERAL EQUILIBRIUM

CONSUMER. To close the model, we introduce a representative consumer who has constant elasticity of substitution preferences over N crops with elasticity of substitution σ and crop-specific taste weights φ_i . The consumer purchases quantity C_i of each crop from farms at price p_i . The consumer owns flexible inputs and rents out the endowment X_g^{agg} of each input g to the farms at rental rate r_g . The consumer also owns all farms and receives their profits as dividends Π .

The consumer's problem is:

$$\max_{\{C_i\}_{i=1}^N} \left(\sum_i \varphi_i C_i^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \quad (5)$$

s.t.

$$\sum_i p_i C_i = \sum_g r_g X_g^{agg} + \Pi \quad (6)$$

For conciseness, we adopt the following formulation of the dividend Π :

$$\Pi = \sum_f \left[\sum_{i=1}^N p_i y_{f,i} - \sum_{g=1}^G r_g \sum_{i=1}^N x_{f,g,i} \right]$$

In this formulation, farmers act *as if* they faced distortions $\tau_{f,g}$ and $\tau_{f,g,i}$, fixed cost ω , and additional concavity η (since these show up in farmers' maximization problems), but these are ultimately non-monetary frictions and obstacles and so do not show up in the dividends Π that farmers send to the consumer. An exactly equivalent formulation would be to interpret distortions, fixed costs, and additional concavity as monetary taxes or subsidies administered by the consumer: in this case, they would be included in Π and would likewise show up (with an opposite sign) in the consumer's budget constraint as government revenue or expense.¹²

MARKET CLEARING. Crop prices p_i and input prices r_g need to be such that all crops and all inputs markets clear:

$$C_i = \sum_f y_{f,i} \quad \forall i \quad (7)$$

$$\sum_f \sum_i x_{f,g,i} = X_g^{agg} \quad \forall g \quad (8)$$

Land needs to be distributed between farms in such a way that land used in cultivation equals the total amount of land available:

$$\sum_f \sum_i l_{f,i} = L^{agg} \quad (9)$$

The solution and the computational algorithm are discussed in Appendix [C.2](#).

12. Extending this logic, distortion terms τ can be interpreted as a mixture of non-monetary frictions and monetary government taxes/subsidies. In this case, a fraction (representing the tax/subsidy portion) of each τ would show up in Π and the consumer's budget constraint. The choice between all these interpretations is completely arbitrary as they collapse to identical equilibrium conditions.

5.3 MAPPING OBSERVABLES TO FUNDAMENTALS

For the model to be useful for quantifying the cost of misallocation in the data, unobserved distortions τ need to be estimable from observed characteristics. In the class of misallocation models following [Hsieh and Klenow \(2009\)](#), the fundamental distortions faced by each firm can be mapped to marginal revenue products of inputs chosen by the firm. With the assumed production structure, unobserved marginal revenue products can be extracted from observed inputs and outputs.

This result extends to our multi-product model as well, for both flexible inputs g and the land input l :

$$r_g \tau_{f,g} \tau_{f,g,i} = \alpha_{g,i} \eta \frac{(p_i y_{f,i})^\eta}{x_{f,g,i}} \quad (10)$$

$$\lambda_f \tau_{f,l,i} = \gamma_i \eta \frac{(p_i y_{f,i})^\eta}{l_{f,i}} \quad (11)$$

While only the total cost of the input can be extracted from the data in this way, the split of the total costs $r_g \tau_{f,g} \tau_{f,g,i}$ and $\lambda_f \tau_{f,l,i}$ into their components is arbitrary from the point of view of the farm. All that matters for quantifying misallocation is the *dispersion* in these costs, which is driven entirely by the dispersion in distortion τ terms. The average *level* of each distortion is then split from the level of market rental rates r_g and the average land shadow cost by the general equilibrium market clearing conditions. Furthermore, goods taste weights φ_i are picked to ensure that the current allocation is consistent with a general equilibrium: these details are discussed in [Appendix C.2](#).

The productivity of the farm in each grown crop $z_{f,i}$ can be extracted from the data directly using the assumed production function, exploiting the availability of physical input and output quantities in the data:

$$z_{f,i} = \frac{y_{f,i}}{l_{f,i}^{\gamma_i} \Pi_g \left(x_{f,g,i}^{\alpha_{g,i}} \right)} \quad (12)$$

SEASONS. As discussed in [Section 3](#), viewing farms' production at the annual level may be deceptive as farmers tend to produce different products in different seasons. Bundling production across seasons may lead to nonsensical counterfactual predictions in the model like the model farm reallocating three times its total amount

of land (summed across three seasons) to a single crop that's only ever grown in the most productive of the three seasons. Therefore, we map each model farm f not to an empirically observed farm, but to a farm-season combination. This ensures that different seasons have distinct productivity distributions. Section 6.1 will discuss the treatment of seasons at the aggregate level.

5.4 CALIBRATION

ADDITIONAL CONCAVITY PARAMETER η . Farm-crop revenue can be written as:

$$p_i y_{f,i} = \underbrace{\left(\left(\frac{1}{\lambda_f \tau_{f,l,i}} \right)^{\gamma_i} \Pi_g \left(\frac{1}{\tau_{f,g} \tau_{f,g,i}} \right)^{\alpha_{g,i}} \right)^{\frac{1}{1-\eta(\sum_g \alpha_{g,i} + \gamma_i)}}}_{\text{composite distortion, } dist_{f,i}} \cdot \underbrace{\left(p_i z_{f,i} \gamma_i^{\gamma_i} \eta^{\sum_g \alpha_{g,i} + \gamma_i} \Pi \left(\frac{\alpha_{g,i}}{r_g} \right)^{\alpha_{g,i}} \right)^{\frac{1}{1-\eta(\sum_g \alpha_{g,i} + \gamma_i)}}}_{\text{"objective" factors}} \quad (13)$$

where the composite distortion term summarizes the effects of distortions on revenue and the “objective factors” term captures effects driven by productivity, production parameters, and prices.

As Section 5.3 above discussed, the model can rationalize any observed heterogeneity across and within farms using τ distortions. However, the variance of distortions needed to achieve this (summarized in the variance of the log $dist_{f,i}$ term) depends on the chosen additional concavity parameter η , as Figure 6 illustrates. When η is low, the degree of concavity in farm-crop revenue is high: in the frictionless optimum, farm-crop revenues are distributed comparatively uniformly, implying a uniform size distribution across farms and significant crop mixing within farms. The current observed allocation is highly dispersed in comparison, and the model rationalizes it with extreme frictions, producing a high variance of $dist_{f,i}$. When η is high, on the other hand, there is little concavity: in the frictionless optimum, farm-crop revenues are dispersed, implying high dispersion in farm sizes and significant crop specialization within farms. The current observed allocation is highly uniform in comparison, and the model rationalizes it with extreme frictions yet again. Somewhere in between is an intermediate value of η that allows the model to reproduce the observed level of dispersion in farm-crop revenues with the most conservative

possible distribution of distortions.¹³ Upon estimation, the value that minimizes the variance of $\log dist_{f,i}$ in Equation 13 is $\eta = 0.63$, and this is the value we elect for our headline calibration.

Misallocation estimates in general are sensitive to the degree of concavity in the model, whether that concavity comes from monopolistic competition (as in Hsieh and Klenow 2009) or decreasing returns to scale in production (as in Chen, Restuccia, and Santaaulalia-Llopis 2022). By introducing a flexible parameter η , we are able to explicitly explore the dependence of our estimates on this parameter (as sensitivity analysis will do in Section 6.6) and to intentionally pick a conservative estimate for the headline calibration.

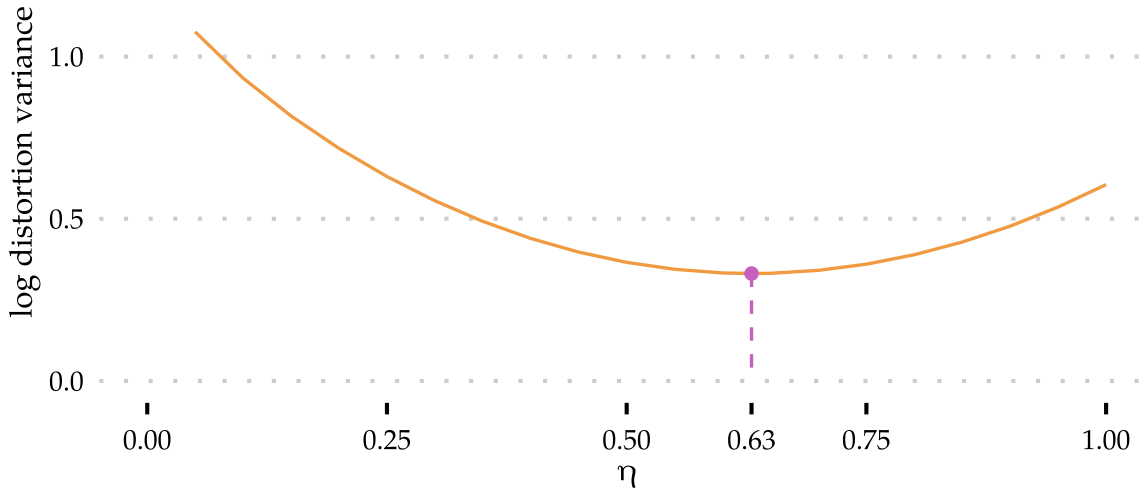


FIGURE 6: Variance of log extracted composite distortion term vs additional concavity parameter η

Note. The figure visualizes the variance of $\log dist_{f,i}$ from Equation 13 at different values of η . The highlighted value of η minimizes the variance.

ELASTICITY OF SUBSTITUTION σ . Representative consumer's demand for product i is

13. Distortions are demeaned within the season.

$$C_i = \left(\frac{\varphi_i}{p_i} \right)^\sigma \frac{\sum_g r_g X_g^{agg} + \Pi}{\sum_j \varphi_j^\sigma p_j^{1-\sigma}} \quad (14)$$

The log of product i 's share in consumption expenditure can then be written as

$$\log \left(\frac{p_i C_i}{\sum_j p_j C_j} \right) = -\log \left(\sum_j \varphi_j^\sigma p_j^{1-\sigma} \right) + (1 - \sigma) \log p_i + \sigma \log \varphi_i \quad (15)$$

Adapting the expression for household-level consumption data with households indexed by h and denoting crop i 's share in h 's expenditure with $s_{h,i}$, the expression can be written as a regression:

$$\log s_{h,i} = \beta_0 + \beta_1 \log p_{h,i} + \gamma_i + \varepsilon_{h,i} \quad (16)$$

Then the elasticity of substitution can be extracted from the coefficient on log price: $\sigma = 1 - \beta_1$. Furthermore, product taste weights can be extracted (up to a normalization) from the estimated product fixed effects: $\varphi_i = \exp \left(\frac{\gamma_i}{\sigma} \right)$.

Estimating Equation 16 as-is will produce biased estimates of β_1 due to potential measurement errors in price creating a mechanical correlation between the left- and the right-hand-side of the equation, and because some unobservables at the household or village level could also be correlated with both crop prices and expenditure shares (Sotelo 2020; Kebede 2020). Therefore, we use an instrumental variable approach to address these concerns, instrumenting for crop prices using the interaction of village-level measures of elevation and ruggedness with rainfall, as well as the availability of pucca roads (all-weather roads). The rationale for using these instruments is that villages with high ruggedness may experience lower agricultural productivity when faced with negative rainfall shocks, consequently leading to higher prices. Similarly, the availability of roads will affect the shadow prices of inputs and hence affect agricultural production. Villages with bad road connectivity may experience higher labor prices and other input prices, raising the cost of crop production and, therefore, negatively affecting the prices of crops. Using these instruments, we estimate the equation 16.

Results of estimation are reported in Appendix Table A.2. We find that the F-statistic for the first stage is 230, which establishes the instrument's relevance. The estimation yields $\beta_1 = -0.7$, implying $\sigma = 1.7$. This is in between the estimates obtained in a similar setting by Kebede (2020) for Ethiopia, 1.3, and by Sotelo (2020)

for Peru, 2.4.

6 QUANTITATIVE RESULTS

To estimate the aggregate cost of misallocation, we conduct counterfactual reallocation exercises in the model. Each exercise reduces or completely removes distortions between farms, computes their optimal counterfactual outputs, and aggregates the output to obtain a counterfactual reallocation output gain. The % difference in output between the counterfactual exercise and the currently observed one constitutes the headline measure of the aggregate cost of misallocation: it captures the magnitude of output being lost due to the market frictions currently present in the economy.

Note that the aggregate input quantities are not changed in the reallocation exercises: only the distortions are. Therefore, any aggregate output gain from reallocation is due purely to the increase in aggregate TFP. Furthermore, because the idiosyncratic farm-crop-level TFPs are not changed in the counterfactual either, the increase in aggregate TFP is driven not by any increase in farm-level TFPs but rather by an improved allocation of existing inputs to the more productive farms.

6.1 BENCHMARK REALLOCATION EXERCISE

In our benchmark reallocation exercise, we equalize all farm-level distortions while keeping the crop set of each farm restricted to what it is observed growing in the data. Furthermore, we leave farm-crop-level distortions intact to preserve idiosyncratic product choice motives.

EQUALIZING DISTORTIONS. We impose equality of farm-input-level distortions: $\tau_{f,g} = \bar{\tau}_g \ \forall g, \lambda_f = \bar{\lambda}$. This equalization effectively makes input markets frictionless across farms. The levels of average distortions are fixed by the general equilibrium conditions. We leave the farm-input-crop-level distortions $\tau_{f,g,i}, \tau_{f,l,i}$ at their current levels to allow the model to capture idiosyncratic product choice motives, as will be discussed below.¹⁴

14. [Aragón, Restuccia, and Rud \(2022\)](#) argue that estimates of agricultural misallocation are more reliable when the unit of analysis is a farm rather than a plot. We rely on plot-level data for production function identification, but in our misallocation exercise, we use variation across plots within a farm only when they produce different crops. Furthermore, leaving farm-input-crop-level distortions intact ensures that any remaining unexplained variation across plots is not interpreted as a source of misallocation, effectively making the exercise capture misallocation between farms and crops, but not

KEEPING CROP SETS FIXED. In this exercise, farmers will change the total quantity purchased of each input as well as the allocation of inputs between the crops they grow. However, farmers are not allowed to change the set of crops they are growing: if a particular farm is observed growing only rice and wheat in the data, it will not be allowed to start growing oilseeds in the counterfactual. This restriction permits us to only use productivities and distortions that are extracted directly from the data: we don't need to make any assumptions on what each farm's counterfactual productivity or farm-crop distortion in an unobserved crop would be.

PRESERVING IDIOSYNCRATIC PRODUCT CHOICE MOTIVES. The model of a profit-maximizing farm is a simplification. Real-world farmers may choose to grow a particular crop for reasons other than pure profit maximization based on observed total factor productivity and input costs. For instance, semi-subsistence farmers may prefer a crop that is less profitable if sold on the market but serves their family's dietary needs better. The model is blind to these motives and would explain this observed preference for a non-profit-maximizing crop with a combination of an inefficient crop set and farm-input-crop-level distortions $\tau_{f,g,i}$ and $\tau_{f,l,i}$ making inputs relatively cheaper for the subsistence crop. Therefore, allowing farmers to change crop sets and equalize farm-input-crop-level distortions may result in the model overstating the true costs of misallocation as it would overstate the degree to which farmers are likely to switch to profit-maximizing crops. Avoiding this is the main reason we elect not to equalize farm-input-crop-level distortions and the second reason we keep crop sets fixed in the benchmark counterfactual exercises. Even though the model does not explicitly include subsistence and other idiosyncratic product choice motives, making these two restrictions makes the counterfactual reallocation exercise only capture the output gain from efficient input markets allowing farmers to more efficiently grow whatever they currently prefer, without making unrealistic predictions on how they could "improve" their existing product preferences.

SEASONS. The treatment of seasons in the model is irrelevant for its ability to reproduce the observed allocation but becomes important for counterfactual exercises. If model farms f , mapped to farm-seasons in the data, are all bundled together into the same economy, a naive counterfactual reallocation may feature double-booking of inputs: land and labor inputs from less productive seasons may be reallocated

between plots.

to the more productive one, implying that a single physical piece of land may be allocated to three different uses in a single season and to no uses in the other two seasons. To preclude this nonsensical possibility, we split the model economy into three sub-economies by season, and conduct counterfactual reallocations only within each season: this ensures that no inputs are reallocated across seasons. The aggregate counterfactual output in each exercise is then a summation of three seasonal outputs.

COMPARING MULTI-PRODUCT AND SINGLE-PRODUCT MODELS. To understand the effect of the product choice channel on aggregate productivity, we compare the estimated reallocation gain between the multi-product farm model and a traditional single-product farm model. The single-product farm model uses inputs and outputs measured at the farm level rather than the farm-crop level. Consequently, the single-product model uses a single production function with estimated “aggregate” elasticities from Figure 3.

6.2 COST OF MISALLOCATION

BENCHMARK REALLOCATION. The benchmark row of Table 2 shows that equalizing input-market distortions across farms raises aggregate agricultural output by 137% in the multi-product model.¹⁵ The single-product model predicts a gain of 167%. It therefore *overstates* the benchmark cost of misallocation by 22%, or 30 percentage points of current output.

“BEST STATE” REALLOCATION. A blanket removal of distortions may be unattainable, and some measured dispersion may reflect model misspecification rather than frictions whose removal would raise productivity.¹⁶ We therefore consider a more modest policy that reduces dispersion to the level of the least-distorted Indian state. We rank states by the variance of the log composite distortion $dist_{f,i}$, defined in Section 5.4, and identify Tamil Nadu as the least distorted. We then rescale each type of

15. Large costs of agricultural misallocation have also been estimated in other developing economies, notably by [Chen, Restuccia, and Santaella-Llopis \(2022\)](#).

16. Potential sources include mismeasurement, late-season productivity shocks, and dynamic input adjustment costs ([Bils, Klenow, and Ruane 2021](#); [Gollin and Udry 2021](#); [Asker, Collard-Wexler, and De Loecker 2014](#)).

TABLE 2: Reallocation exercises

	Gain	Gain (1 product)	Error (1 product)
benchmark	137%	167 %	22%
best state	34%	43 %	27%
land only	27%	33 %	23%
complete	150%	167 %	11%

Note. Gains are percentage increases in aggregate output relative to the observed allocation. “Error (1 product)” is the difference between the single-product and multi-product gains divided by the multi-product gain; positive values indicate overstatement. The benchmark equalizes cross-farm input distortions while retaining within-farm wedges. “Best state” rescales distortion variances to those of Tamil Nadu. “Land only” equalizes cross-farm land shadow costs. “Complete” also equalizes within-farm wedges. Crop sets and aggregate input quantities are fixed, and inputs are reallocated only within seasons.

distortion so that its population log variance matches that in Tamil Nadu.¹⁷ This policy raises output by 34% in the multi-product model and 43% in the single-product model. The single-product estimate overstates the gain by 27%, or 9 percentage points of current output.

ALTERNATIVE REALLOCATIONS. Our benchmark reallocation removes estimated frictions in all three modeled input markets: land, labor, and intermediates. The agricultural misallocation literature, however, has mainly focused on institutional frictions in the land market. In the alternative “land only” reallocation, we remove land frictions between farms (equalize λ_f) but keep their labor and intermediates frictions in place ($\tau_{f,g}$ stay at current levels). The expected reallocation gain in this case is far more modest but still sizable: 27% of current agricultural output. The single-product model estimates a slightly higher reallocation gain of 33%. Note that [Bolhuis, Rachapalli, and Restuccia \(2021\)](#) arrived at a very similar gain of 38% from efficiently reallocating land in India, using a conventional single-product model.

The “land only” results are instructive also because of the gap between them and the benchmark exercise: efficiently reallocating all three inputs is much more effective

17. Variances are computed and equalized separately for five types of distortions: $\hat{\lambda}_f$, $\tau_{f,g}$ (separately for each $g = 1, 2$), $\tau_{f,g,i}$, $\tau_{f,l,i}$. Letting $V_{d,s}$ be the variance of log distortion of type d in the least distorted state and $\hat{V}_d$ be the corresponding population variance, each distortion is rescaled to $\tau_{d,f}^{new} = \exp\left(\sqrt{\frac{V_{d,s}}{\hat{V}_d}} \cdot \log \tau_{d,f}\right)$.

than reallocating land alone, implying that labor and input market frictions impose a significant drag on Indian agricultural productivity.¹⁸

The benchmark reallocation removes frictions between farms but keeps farm-crop-level distortions intact in order to preserve any idiosyncratic product choice motives that the model does not explicitly capture. However, some of the farm-crop-level $\tau_{f,g,i}$ distortions are likely to reflect malign frictions rather than benign unmodeled motives. To place an upper bound on their importance, we conduct an alternative “complete” reallocation that equalizes these farm-crop-level frictions. Their contribution is relatively minor, increasing the reallocation gain from the 137% of the benchmark exercise to 150%. Because the single-product model is blind to within-farm crop-level distortions, its gain is unaffected.

6.3 DECOMPOSITION STRATEGY

We decompose the gap between the models by removing one feature at a time from the multi-product model until it becomes the single-product model. Each intermediate model specifies a production technology, extracts productivity and distortions from the data, and predicts the effects of a common reallocation policy. All six models fit the same observed allocation.

COMMON DATA AND POLICY. Let Y_0 denote the observed aggregate output. For model Π_k , let Y_k be counterfactual output and define its reallocation gain as

$$G_k = \frac{Y_k}{Y_0} - 1. \quad (17)$$

The endpoints are the multi-product model $\Pi_0 = M$ and the single-product model $\Pi_5 = S$. Every intermediate model uses the same farm-crop observations. The crop-resolved models reproduce the observed inputs and outputs of each farm-crop pair, and the single-product model reproduces the corresponding farm totals.

Changing a production function changes both productivity and the distortions inferred from observed input-output choices. We therefore re-extract those objects under each technology before applying the counterfactual policy.

18. See [Andrew D. Foster and Mark R. Rosenzweig \(2022\)](#) for an exploration of labor market frictions in India.

SEQUENCE OF MODELS. Table 3 describes the sequence. We remove, in order: returns-to-scale heterogeneity, factor-intensity heterogeneity, within-farm input reallocation, product-specific demand, and finally product-specific production observations.

TABLE 3: Models used in the mechanism decomposition

Model	Specification and change from preceding model
$\Pi_0 = M$	Multi-product model.
Π_1	Common returns to scale.
Π_2	Common factor intensities.
Π_3	Each input's crop shares are fixed at their current Π_2 values.
Π_4	Homogeneous-good demand replaces CES demand.
$\Pi_5 = S$	Single-product model. Crop observations are aggregated to farm totals.

Note. Changes are cumulative. Every model reproduces common current output Y_0 from the same retained observations and applies the same counterfactual distortion policy. Farms can only operate crops observed in the current allocation.

To separate the two technology changes, collect the land and flexible-input elasticities γ_i and $\alpha_{g,i}$ as $\beta_{h,i}$, where h indexes all inputs. Crop i 's returns to scale are $\rho_i = \sum_h \beta_{h,i}$, and its normalized factor intensities are $\theta_{h,i} = \beta_{h,i}/\rho_i$. Denote the estimated pooled elasticities by $\bar{\beta}_h$, with $\bar{\rho} = \sum_h \bar{\beta}_h$ and $\bar{\theta}_h = \bar{\beta}_h/\bar{\rho}$. Model Π_1 assigns elasticities $\bar{\rho}\theta_{h,i}$: proportional expansion has the same return in every crop, but relative input requirements remain crop-specific. Model Π_2 instead assigns $\bar{\rho}\bar{\theta}_h = \bar{\beta}_h$ to every crop. Both steps retain crop-level demand and allow farms to change input allocations across their existing crops.

Model Π_3 freezes each input's crop shares at their solved current Π_2 values. For flexible input g , the restriction is $x_{f,g,i} = s_{f,g,i}^0 X_{f,g}$, with the analogous restriction for land. Farms can change input totals, and inputs can move between farms. Because technologies have already been pooled, this step measures the remaining adjustment to crop productivities, prices, and retained within-farm wedges. Technology-driven crop adjustment is already part of the first two links. Appendix D describes the restricted farm problem.

Model Π_4 keeps the producers and fixed shares of Π_3 , but treats all crops as one homogeneous consumption good. All crop prices equal one, and aggregate output is $Y = \sum_{f,i} y_{f,i}$. This step removes crop-specific relative-price feedback without changing the placement of η or the crop-level productivity and distortion objects. Model

Π_5 then aggregates crop observations to farm totals and re-extracts single-product productivity and distortions. Producer aggregation changes the revenue term from $\sum_i (p_i y_{f,i})^\eta$ to $(\sum_i p_i y_{f,i})^\eta$ and embeds crop-level latent objects in farm-level measures. The final link therefore includes both curvature placement and the remapping of productivity and distortions. Appendix E gives the common-good equilibrium and producer-aggregation algebra.

CONTRIBUTIONS AND INTERPRETATION. Define the five contributions as

$$\begin{aligned} C_{RS} &= G_1 - G_0, & C_{INT} &= G_2 - G_1, \\ C_{MIX} &= G_3 - G_2, & C_D &= G_4 - G_3, \\ C_P &= G_5 - G_4. \end{aligned} \tag{18}$$

Their sum is the single-product error:

$$E = G^S - G^M = C_{RS} + C_{INT} + C_{MIX} + C_D + C_P. \tag{19}$$

A positive contribution means that removing the named feature raises the estimated gain along this sequence.

6.4 DECOMPOSITION RESULTS

Demand and producer aggregation account for most of the benchmark discrepancy. Figure 7 traces the five contributions from the multi-product estimate to the single-product estimate, expressing every step as a change in the cumulative error. Table 4 compares the benchmark with complete and best-state reallocations.

BENCHMARK. Equalizing returns to scale lowers the estimated gain by 1.4 percentage points; pooling factor intensities then raises it by 1.8 points. Freezing within-farm input shares adds another 0.7 points. Together these three changes account for only 1.1 points of the 30.1-point discrepancy. The small net contribution does not imply that crop technologies or adjustment never matter: each link measures their effect under the benchmark policy and after the preceding changes.

Demand aggregation contributes 24.7 points. Under CES demand, expansion of a crop changes its relative price and the consumption value of additional output. Treating all crops as a common good removes that feedback and substantially raises

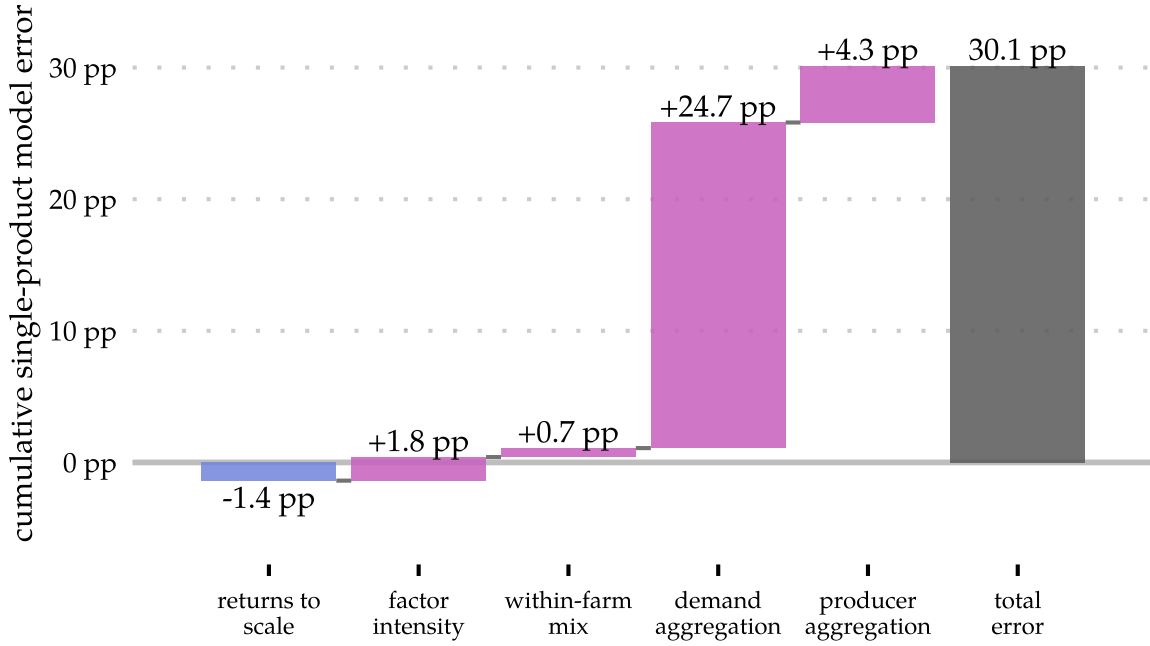


FIGURE 7: Decomposition of the benchmark single-product model error

Note. The waterfall starts at zero error relative to the multi-product gain of 137%. Each step removes the indicated feature in the order of Table 3. The five contributions sum to the 30.1 percentage-point error, giving a single-product gain of 167%. The final gray bar shows this total error. All plotted quantities are percentage points of current output.

TABLE 4: Mechanism contributions across reallocation policies

Feature	Benchmark	Complete	Best state
Returns-to-scale heterogeneity	-1.38	+1.83	-1.70
Factor-intensity heterogeneity	+1.79	+0.74	+3.63
Within-farm reallocation margin	+0.68	-14.48	+4.09
Demand aggregation	+24.74	+24.74	+3.68
Producer aggregation	+4.32	+4.32	-0.44
Total single-product error	+30.14	+17.14	+9.25

Note. Entries are percentage points of current output. Each row is the change in the estimated reallocation gain when the named feature is removed after the preceding features. Positive entries raise the single-product estimate relative to the multi-product estimate. Each column's five contributions sum to the total error before rounding. Policy definitions are those of Table 2.

the predicted gain in this exercise. Aggregating producers adds 4.3 points through the placement of crop-level concavity and the re-extraction of farm-level objects. Together, demand and producer aggregation account for 29.1 points, or 96.4% of the gap between the two models. This share concerns the discrepancy between estimates, not the total gain from removing cross-farm frictions.

ALTERNATIVE POLICIES. Equalizing within-farm wedges changes the role of crop adjustment. In the complete reallocation, the returns-to-scale and factor-intensity links contribute 1.8 and 0.7 points, respectively, while freezing within-farm shares lowers the gain by 14.5 points. Flexible crop adjustment therefore raises output substantially in this exercise. The demand and producer contributions remain 24.7 and 4.3 points, leaving a smaller total error of 17.1 points.

Under the best-state policy, returns-to-scale equalization lowers the gain by 1.7 points, whereas intensity pooling adds 3.6 points and freezing shares adds 4.1 points. Demand aggregation contributes 3.7 points and producer aggregation offsets 0.4 points, producing a total error of 9.3 points. The discrepancy is spread more evenly across the model features than in the benchmark.

6.5 APPARENT MISALLOCATION WITHOUT CROSS-FARM FRICTIONS

The single-product model predicts a further 12.3% output gain even after cross-farm frictions have been equalized in the multi-product benchmark. To obtain this diagnostic, we treat the multi-product counterfactual as new data: we sum each farm's crop inputs and outputs, re-extract productivity and distortions using the single-product model, and conduct its reallocation exercise. Differences in product composition can still generate farm-level input ratios that the pooled production function interprets as distortions.

The apparent gain remains 11.5% when the source allocation also equalizes all within-farm wedges. The single-product procedure can thus report additional misallocation after every modeled input wedge has been equalized, simply by misinterpreting heterogenous input choice due to product heterogeneity as evidence of frictions.

6.6 SENSITIVITY AND MARKET BOUNDARIES

PARAMETER SENSITIVITY. Figure 8 varies the additional concavity parameter η and the elasticity of substitution between crops σ around the benchmark calibration. Panel (A) shows that reallocation gains are sensitive to these parameters. With less concavity (higher η or σ), productive farms can expand more when distortions are removed. This sensitivity motivates our choice of η to minimize the variance of extracted composite distortions, as described in Section 5.4.

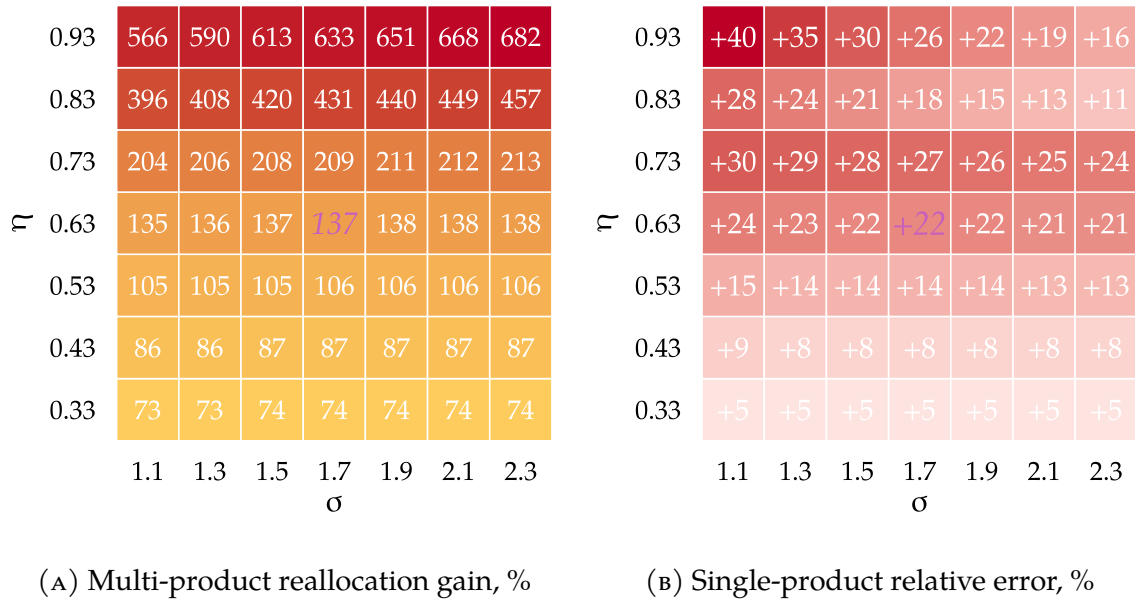


FIGURE 8: Reallocation gain and model error under alternative parameterizations

Note. Panel (A) reports the percentage increase in aggregate output from the benchmark distortion policy. Panel (B) reports the single-product model's relative overstatement. The central highlighted values use the calibrated $\sigma = 1.7$ and $\eta = 0.63$.

Panel (B) shows overstatement at all 49 parameter combinations, ranging from 5% to 40%, compared with 22% at the benchmark. Thus the benchmark direction is robust within this grid, but the size of the correction varies substantially.

STRENGTH OF REALLOCATION. Does the decomposition of reallocation gain change with the strength of reallocation? We progressively compress the variance of measured distortions, applying the same rule to all six intermediate models at each point. Figure 9 reports the composition of the resulting model error. Each contribution is

divided by the total error at that point, so the five shares sum to 100%. The total error is positive throughout the grid and falls from 29.87 percentage points under the strongest reallocation to 0.02 points under the weakest. Demand aggregation remains an important mechanism throughout the range of reallocation strengths. Returns to scale heterogeneity and producer aggregation are equally important at low reallocation strengths, but their relative importance diminishes as reallocation strengthens.

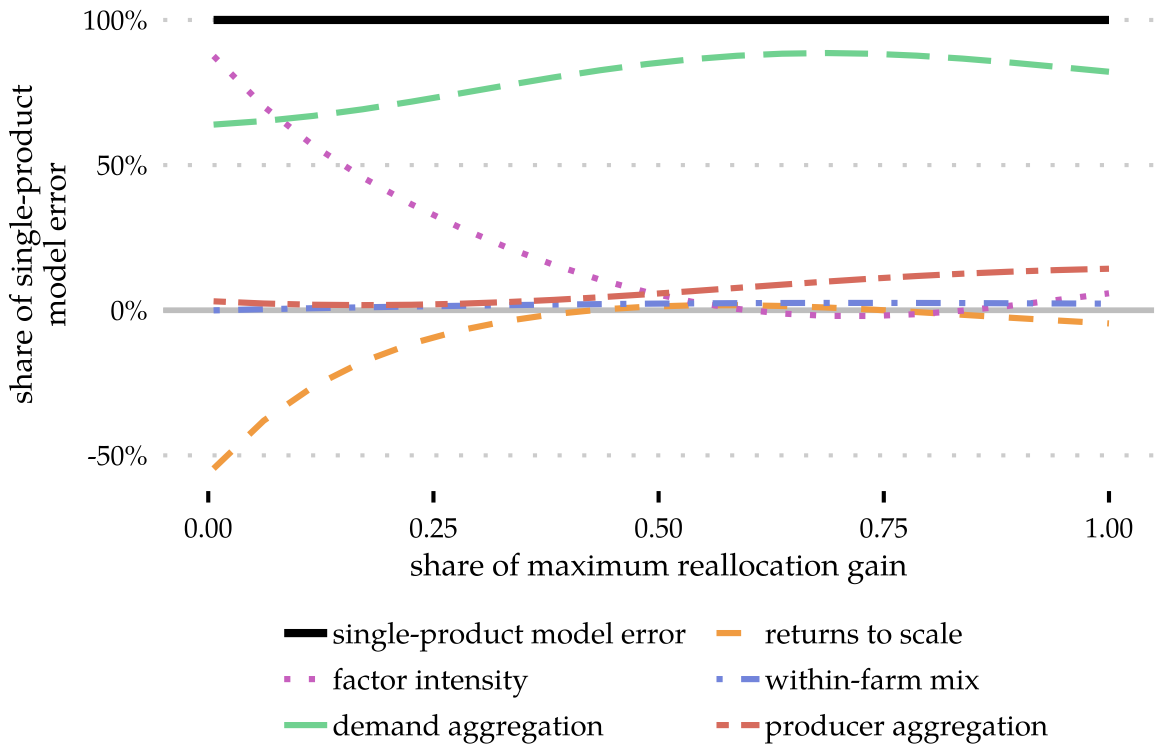


FIGURE 9: Composition of the single-product model error across reallocation strengths

Note. The horizontal axis is the multi-product gain divided by the largest multi-product gain on the distortion-variance grid. The five shares sum to the black 100% reference line. A negative share offsets positive contributions. Each point uses the sequence and common-policy comparison in Section 6.3.

ROLE OF SEASONS AND STATES. Restricting reallocation to occur within each state lowers the multi-product benchmark gain to 101%, compared with 112% in the single-product model (Appendix Table A.4). The single-product estimate overstates this gain by 11%. Each season-state economy includes only crops observed there, preserving the same local crop support throughout the comparison.

Allowing inputs to move across seasons instead raises the multi-product benchmark gain to 148% and the single-product gain to 211%, an overstatement of 43% (Appendix Table A.5). This exercise illustrates why the seasonal restrictions matter: a gain obtained by assigning land from several seasons to one season need not be physically feasible. Our preferred specification retains separate seasonal markets.

7 CONCLUSION

Misallocation of inputs between firms imposes a significant drag on the productivity of developing countries. The cost of misallocation in agriculture is especially large and contributes to explaining the enormous agricultural productivity gap between developed and low-income countries.

Estimates of misallocation in non-agricultural sectors have been limited by the treatment of all firms as producing only a single good or multiple homogeneous goods. Estimates of agricultural misallocation have been further limited by assuming that all farms produce the same agricultural product. We use Indian farm-level data to estimate production functions for individual crops and find that their input elasticities and returns to scale are meaningfully heterogeneous for the purposes of estimating misallocation.

Our model of multi-product firms implies a 137% productivity gain from equalizing cross-farm input-market frictions in India, compared with 167% in the conventional single-product model. The single-product estimate overstates the gain by 22%. We account for the gap through five sequential changes: pooling returns to scale, pooling factor intensities, freezing within-farm input shares, aggregating demand, and aggregating producers. The five contributions add exactly to the difference between models. In the benchmark, demand and producer aggregation account for 96% of that difference.

Product aggregation also generates apparent misallocation after cross-farm input-market frictions have been removed. Applied to the multi-product benchmark counterfactual, the single-product measurement and reallocation procedure predicts an additional 12% gain relative to that counterfactual's output. The predicted gain remains 12% when within-farm wedges are equalized too. Farm-level input ratios can therefore indicate distortions in the pooled model even when they reflect crop composition and technology in the multi-product economy.

We conducted our analysis on Indian farm-level data, since agricultural micro-

level data provides several advantages in exploring the interplay of product heterogeneity and misallocation. However, the model developed in the paper—and the mechanisms it illustrates—would apply equally to non-agricultural firms. Future research can extend this analysis from agriculture to non-agriculture by applying the model to firm-product-level datasets.

Another future direction is to consider the role of dynamics in the interaction of product choice and misallocation.¹⁹ Crop rotation can lead to input choices that appear inefficient in the short run but preserve the long-run productivity of land. Moreover, the heterogeneous suitability of different crops to different seasons can produce—when input rental markets are underdeveloped—input choices that appear inefficient in one season but are made to have the input in place for the next season. Capturing product choice complementarities across seasons and even years empirically and theoretically is thus a fruitful avenue of further research.

19. Dynamic choices have been shown to be important for misallocation using single-product models by, for instance, [Asker, Collard-Wexler, and De Loecker \(2014\)](#) and [Kehrig and Vincent \(2019\)](#).

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APPENDIX

A ADDITIONAL FIGURES AND TABLES

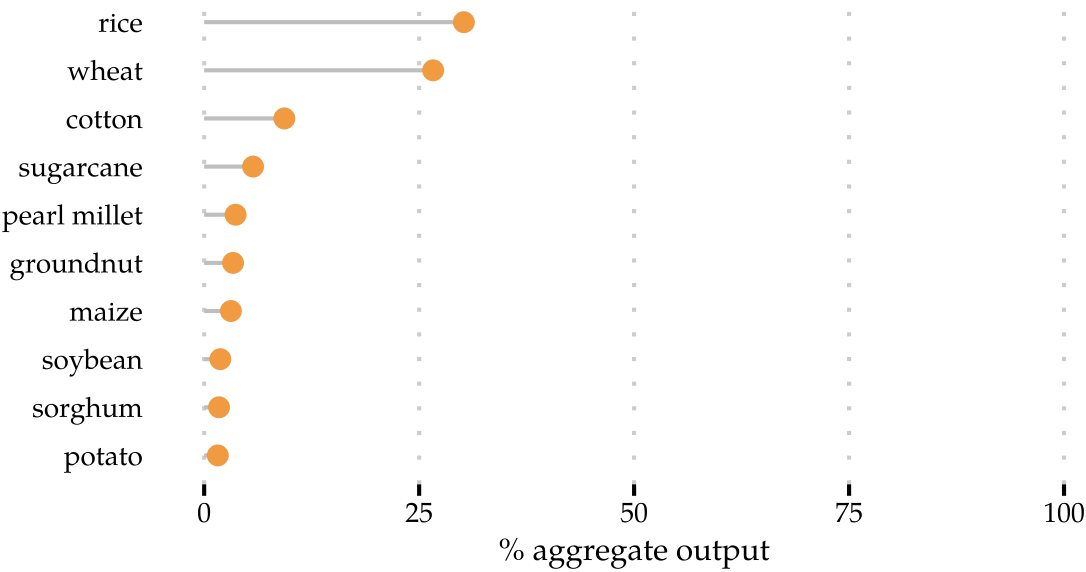


FIGURE A.1: Top 10 crops by % of aggregate output

TABLE A.1: Production function estimates

	(1)	(2)	(3)	(4)	(5)	(6)
	Aggregate	Rice	Wheat	Other Cereals	Pulses	Veg, Frt, Oil
Land	0.427 (0.031)	0.511 (0.021)	0.414 (0.032)	0.407 (0.034)	0.379 (0.060)	0.330 (0.035)
Labor	0.189 (0.031)	0.161 (0.037)	0.122 (0.036)	0.248 (0.058)	0.316 (0.074)	0.430 (0.064)
Intermediates	0.351 (0.041)	0.307 (0.033)	0.517 (0.040)	0.245 (0.046)	0.255 (0.076)	0.127 (0.052)
Observations	14,705	4,807	3,566	2,779	1,128	2,338
R^2	0.624	0.742	0.713	0.590	0.417	0.572
Village FEs	Y	Y	Y	Y	Y	Y
Season FEs	Y	Y	Y	Y	Y	Y
First Stage: F statistics						
Land	77.0	62.0	40.3	37.8	15.7	19.3
Labor	49.3	34.7	17.7	25.2	12.9	14.8
Intermediates	35.8	31.7	21.5	19.9	8.9	11.8
K-Paap Wald F statistic	51.1	40.4	16.0	30.8	12.4	12.7

Note. Each column represents a separate regression. We use 2SLS to estimate the coefficients, using instruments described in Section 4.2. Robust standard errors are presented in parentheses. The “Other Cereals” group include barley, maize, jawar, bajra, ragi, and millets. The “Veg, frt, oil” group includes vegetables, fruits, oilseeds, and fiber crops. The “Aggregate” column pools all crops together. In panel B of the table, we report the F-stat from the first stage regression for each input and each regression separately. We also report the joint F-stat at the bottom line of the table.

TABLE A.2: Estimate of σ

	(1) $\log s_{h,i}$	(2) $\log p_{h,i}$
σ	1.699	
$\log p_{h,i}$	-0.699 (0.067)	
Elevation $\times$ rain		-0.004 (0.000)
Ruggedness $\times$ rain		0.016 (0.001)
Pucca roads availability		-0.023 (0.001)
Observations	40,833	40,833
Kleibergen-Paap F stat	230.9	

Note. The table presents the estimation of Equation 16 using 2SLS. Column 1 presents second stage results and column 2 presents first stage results. Elevation, ruggedness, rain, and availability of pucca roads are village-level measures. “Pucca roads” are all-weather roads made of concrete and tar. Standard errors are clustered at the household level. The “ σ ” row shows the value of σ implied by β_1 , the coefficient on $\log p_{h,i}$.

TABLE A.3: Crop list

Rice	Wheat	Other Cereals	Pulses	Oilseeds, Fruits and Vegetables		
		Barley	Black gram	Oilseeds	Vegetables	Fruits /Condiments
		Maize	Green peas	Sesame	Ash gourd	Mango
		Sorghum	Pigeon peas	Groundnut	Beet root	Papaya
		Pearl millet	Horse gram	Castor	Bitter gourd	Grapes
		Finger millet	Cowpea	Sunflower	Bottle gourd	Plum
		Others	Kidney bean	Niger	Eggplant	Cardamom
			Lentil	Soybean	Board bean	Chilli
			Chickpeas	Safflower	Cabbage	Cumin
			Others	Rapseed	Cauliflower	Dill seed
				Linseed	Carrot	Indian mustard
				Others	Potato	Other
					Cucumber	
					Peas	

TABLE A.4: Reallocation exercises: within state

	Gain	Gain (1 product)	Error (1 product)
benchmark	101%	112 %	11%
land only	19%	24 %	25%
complete	112%	112 %	0%

Note. Unlike the main exercises in Table 2, the present exercises only re-allocate inputs within each state: effectively, separate states are treated as separate economies. The nomenclature follows that of Table 2.

TABLE A.5: Reallocation exercises: no split by season

	Gain	Gain (1 product)	Error (1 product)
benchmark	148%	211 %	43%
best state	40%	45 %	12%
land only	27%	33 %	23%
complete	161%	211 %	31%

Note. Unlike the main exercises in Table 2, the present exercises reallocate inputs across seasons: separate seasons are *not* treated as separate economies. The nomenclature follows that of Table 2.

B ADDITIONAL CONCAVITY MECHANISMS

Additional concavity governed by the parameter η , described in Section 5.1, is a reduced-form stand-in for multiple mechanisms that could incentivize farmers to produce multiple crops at once.

In this section, we explore two such mechanisms explicitly. Both generate first-order conditions that are functionally equivalent to those in the original problem. At the same time, different micro-foundations of η may carry with them different natural ways of aggregating crops and clearing markets at the aggregate level, generating slightly different general equilibrium implications.

B.1 SUBSISTENCE

Suppose that instead of operating a profit-maximizing farm, the household is operating a subsistence farm and consuming its output. The farmer is minimizing costs

of providing utility constant elasticity of substitution utility U to the family:

$$U = \left(\sum_i \varphi_i y_{f,i}^\eta \right)^{\frac{1}{\eta}}$$

In this interpretation, the farmer has an incentive to mix crops in order to provide a diverse diet for the family. The resulting first-order conditions are equivalent to those in Section 5.1.

B.2 MARKET POWER

Suppose that instead of selling the harvest at a perfectly competitive market, the farmer has some market power in a monopolistically competitive market. This can be represented by introducing an intermediate crop aggregator operating within each crop i 's market, which buys farms' varieties $y_{f,i}$ of this crop at $p_{f,i}$ and sells the combined i product to the consumer at price P_i . The aggregator uses the following production function:

$$Y_i = \left(\sum_f y_{f,i}^\eta \right)^{\frac{1}{\eta}}$$

This generates downward-sloping demand for farm f 's variety of crop i , creating an incentive for the farmer to mix crops (to avoid excessive price drops in any one crop). The resulting first-order conditions are equivalent to those in Section 5.1

C MODEL SOLUTION AND COMPUTATIONAL ALGORITHM

C.1 SOLVING THE PROBLEM OF A SINGLE FARM

For N products, there are $2^N - 1$ non-empty product sets that each farm could be producing. Let I_f be the set of products in a single such combination, s.t. $y_{f,i} > 0 \forall i \in I_f$ and $y_{f,i} = 0 \forall i \notin I_f$. If $\omega > 0$, the profit-maximizing product set might have $y_{f,i} = 0$ for some crops. Each combination has to be solved separately, and then their profits are compared to pick the best one.

For each product set I_f , the following non-linear equation needs to be solved for

the shadow cost of land λ_f :

$$\sum_{i \in I_f} \lambda_f^{\frac{\eta \sum_g \alpha_{g,i} - 1}{1 - \eta \sum_g \alpha_{g,i} - \eta \gamma_i}} \left((p_i z_{f,i})^\eta \cdot \eta \cdot \Pi_g \left(\frac{\alpha_{g,i}}{r_g \tau_{f,g} \tau_{f,g,i}} \right)^{\eta \alpha_{g,i}} \left(\frac{\gamma_i}{\tau_{l,i}} \right)^{1 - \eta \sum_g \alpha_{g,i}} \right)^{\frac{1}{1 - \eta \sum_g \alpha_{g,i} - \eta \gamma_i}} \tau_{l,i} = L_f \quad (\text{C.1})$$

Each term within the sum is decreasing in λ_f . Using this fact, the following bounds can be derived on λ_f that solves the condition, simplifying the numerical cost. For each crop, define:

$$A_i = \left((p_i z_{f,i})^\eta \cdot \eta \cdot \Pi_g \left(\frac{\alpha_{g,i}}{r_g \tau_{f,g} \tau_{f,g,i}} \right)^{\eta \alpha_{g,i}} \left(\frac{\gamma_i}{\tau_{l,i}} \right)^{1 - \eta \sum_g \alpha_{g,i}} \right)^{\frac{1}{1 - \eta \sum_g \alpha_{g,i} - \eta \gamma_i}} \tau_{l,i} \quad (\text{C.2})$$

Then the upper and lower bounds on λ_f are

$$\lambda_f^{UB} = \max_i \left(\frac{L_f}{|I_f| \cdot A_i} \right)^{\frac{1 - \eta \sum_g \alpha_{g,i} - \eta \gamma_i}{\eta \sum_g \alpha_{g,i} - 1}} \quad (\text{C.3})$$

$$\lambda_f^{LB} = \min_i \left(\frac{L_f}{|I_f| \cdot A_i} \right)^{\frac{1 - \eta \sum_g \alpha_{g,i} - \eta \gamma_i}{\eta \sum_g \alpha_{g,i} - 1}} \quad (\text{C.4})$$

Once the λ_f for this product set is known, input choices for products within I_f can be obtained from:

$$x_{f,h,i} = \frac{\alpha_{h,i}}{r_h \tau_{f,h} \tau_{f,h,i}} \left((p_i z_{f,i})^\eta \cdot \eta \cdot \Pi_g \left(\frac{\alpha_{g,i}}{r_g \tau_{f,g} \tau_{f,g,i}} \right)^{\eta \alpha_{g,i}} \left(\frac{\gamma_i}{\lambda_f \tau_{l,i}} \right)^{\eta \gamma_i} \right)^{\frac{1}{1 - \eta \sum_g \alpha_{g,i} - \eta \gamma_i}} \quad (\text{C.5})$$

$$l_{f,i} = \left((p_i z_{f,i})^\eta \cdot \eta \cdot \Pi_g \left(\frac{\alpha_{g,i}}{r_g \tau_{f,g} \tau_{f,g,i}} \right)^{\eta \alpha_{g,i}} \left(\frac{\gamma_i}{\lambda_f \tau_{l,i}} \right)^{1 - \eta \sum_g \alpha_{g,i}} \right)^{\frac{1}{1 - \eta \sum_g \alpha_{g,i} - \eta \gamma_i}} \quad (\text{C.6})$$

Set $x_{f,h,i} = l_{f,i} = 0 \ \forall i \notin I_f$.

With input choices computed, the profit for this product set can be evaluated using Equation 2. Iterate over $2^N - 1$ possible product sets and choose the product set yielding the highest profit.

C.2 SOLVING FOR THE GENERAL EQUILIBRIUM

Merge consumer's product demands, farms' input demands, farms' optimal output, and market clearing conditions to obtain market clearing conditions in terms of fundamentals, shadow costs of land λ_f , goods prices p_i , inputs prices x_g , and farms' product sets I_f (letting the set of farms producing each product to be denoted by F_i):

$$\sum_f \sum_{i \in I_f} \left((p_i z_{f,i})^\eta \cdot \eta \cdot \Pi_g \left(\frac{\alpha_{g,i}}{r_g \tau_{f,g} \tau_{f,g,i}} \right)^{\eta \alpha_{g,i}} \cdot \left(\frac{\gamma_i}{\lambda_f \tau_{f,l,i}} \right)^{1-\eta \sum_g \alpha_{g,i}} \right)^{\frac{1}{1-\eta \sum_g \alpha_{g,i} - \eta \gamma_i}} = L^{agg} \quad (C.7)$$

$$\sum_f \sum_{i \in I_f} \frac{\alpha_{h,i}}{r_h \tau_{f,h} \tau_{f,h,i}} \left((p_i z_{f,i})^\eta \cdot \eta \cdot \Pi_g \left(\frac{\alpha_{g,i}}{r_g \tau_{f,g} \tau_{f,g,i}} \right)^{\eta \alpha_{g,i}} \cdot \left(\frac{\gamma_i}{\lambda_f \tau_{f,l,i}} \right)^{\eta \gamma_i} \right)^{\frac{1}{1-\eta \sum_g \alpha_{g,i} - \eta \gamma_i}} = X_h^{agg} \quad (C.8)$$

$$p_i = \left(\varphi_i^{-\sigma} \frac{\sum_j \varphi_j^\sigma p_j^{1-\sigma}}{\sum_j p_j \sum_{f \in F_j} y_{f,j}} \cdot \sum_{f \in F_i} \left(\eta^{\sum_g \alpha_{g,i} + \gamma_i} z_{f,i} \Pi_g \left[\left(\frac{\alpha_{g,i}}{r_g \tau_{f,g} \tau_{f,g,i}} \right)^{\alpha_{g,i}} \right] \left(\frac{\gamma_i}{\lambda_f \tau_{f,l,i}} \right)^{\gamma_i} \right)^{\frac{1}{1-\eta \sum_g \alpha_{g,i} - \eta \gamma_i}} \right)^{\frac{1-\eta \sum_g \alpha_{g,i} - \eta \gamma_i}{(1-\eta \sum_g \alpha_{g,i} - \eta \gamma_i)(1-\sigma)-1}} \quad (C.9)$$

The p_i condition can be simplified in two ways, to abstract from its dependence on all other goods' prices p_j . First, the price level can be normalized: $\sum_j \varphi_j^\sigma p_j^{1-\sigma} = 1$. Second, the nominal output can be substituted with a variable Y_{nom} , adding its identity to the list of market-clearing conditions:

$$Y_{nom} = \sum_j p_j \sum_{f \in F_j} y_{f,j} \quad (C.10)$$

SPLITTING DISTORTIONS AND ENSURING THE CURRENT ALLOCATION IS CONSISTENT WITH GENERAL EQUILIBRIUM. Once frictions have been extracted from the data (Section 5.3), additional steps need to be taken to ensure that the model representation of the current allocation is in general equilibrium.

Splitting $r_g \tau_{f,g} \tau_{f,g,i} = mrpg_{f,i}$ into r_g and $\tau_{f,g}$ is arbitrary. To ease the interpretation of frictions, we impose that $\tau_{f,g} \tau_{f,g,i}$ only distort the farm-level input demands and do not affect the aggregate input demand X_g^{agg} . This implies the following market price of g :

$$r_g = \frac{\sum_i \eta \alpha_{g,i} \sum_f (p_i y_{f,i})^\eta}{X_g^{agg}} \quad (C.11)$$

Extending the logic, we impose that $\tau_{f,g,i}$ only distort the within-farm allocation of inputs between crops and do not affect the farm-level input demand, implying the following:

$$\tau_{f,g,i} = \frac{\eta \alpha_{g,i} (p_i y_{f,i})^\eta}{\sum_j \eta \alpha_{g,j} (p_j y_{f,j})^\eta} \frac{\sum_j x_{f,g,j}}{x_{f,g,i}} \quad (C.12)$$

Finally, the unobserved taste weights φ_i need to be consistent with the observed allocation being in general equilibrium. To simplify notation, we map the price-weighted outputs in the data to real output $y_{f,i}$ in the model, implying that all prices in the current model allocation are: $\{p_i\} = 1$. Any such rescaling does not affect the real behavior of the model, neither in the current allocation nor in the counterfactual reallocations, since units of output in different products are arbitrary. This rescaling simplifies extracting the taste weights consistent with the current allocation being in general equilibrium to:

$$\varphi_i = \left(\frac{\sum_f y_{f,i}}{\sum_j \sum_f y_{f,j}} \right)^{\frac{1}{\sigma}} \quad (C.13)$$

Note that this procedure makes extracting taste weights from estimated fixed effects in Equation 16 unnecessary.

WHEN PRODUCT SETS ARE FIXED. Although the current allocation extracted from the data can be easily made to be consistent with the general equilibrium in the model, market-clearing prices have to be solved numerically for any counterfactual reallocation. When product sets are fixed at the currently observed level, the problem of finding market-clearing prices using Equations C.7-C.9 is simplified by the fact that farmers cannot switch their product sets in response to changing prices.

If at least land is being efficiently reallocated between farms (λ_f is equalized between farms), the market-clearing prices can be solved without having to re-solve each individual farm's problem.²⁰ This is the case in the benchmark reallocation exercise discussed in Section 6.1.

In this case, farms' product sets I_f are known from the current allocation. Since land is being reallocated, $\lambda_f = \bar{\lambda} \forall i$. This allows the land market condition C.7, the flexible input markets conditions C.8 and the input market identity C.10 to be solved as a system with $G + 2$ dimensions in $G + 2$ unknowns: $\bar{\lambda}$, $\{r_g\}_g$, and Y_{nom} . On each guess of these variables, goods prices consistent with the guess can be computed from C.9.

WHEN PRODUCT SETS ARE ALLOWED TO CHANGE. When farms are allowed to change their product set I_f from its current level, the need to solve all individual farms' problems on each price guess for new product sets can no longer be avoided.

By analogy with flexible inputs (Section 5.3), let each farm's land shadow cost deviation be fixed at $\hat{\lambda}_f = \lambda_f / \bar{\lambda}$.

The brute-force approach to solving for market-clearing prices is to solve conditions C.7, C.8, C.9 as a system with $G + N + 1$ dimensions in the same number of unknowns: $\bar{\lambda}$, $\{x_g\}_g$, $\{p_i\}_i$. Within each price guess:

1. Solve individual farms' problems. For each farm:
 - (a) For each potential product set I_f , find the land endowment L_f consistent with $\lambda_f = \bar{\lambda} \cdot \hat{\lambda}$ using Equation C.1.
 - (b) Solve the farm's problem for the optimal product set I_f , following Appendix C.1.
2. Evaluate conditions C.7, C.8, C.9.

The brute-force algorithm is guaranteed to find market-clearing prices but is exceptionally computationally costly due to its high-dimensionality and the need to re-solve all farms' problems on each price guess. However, there is a much faster sequential heuristic algorithm:

1. Pick an initial guess of $\bar{\lambda}$, $\{x_g\}_g$, $\{p_i\}_i$.

20. This approach works whether or not any of the flexible input frictions $\tau_{f,g}$, $\tau_{f,g,i}$ are being equalized.

2. Solve individual farms' problems as in the brute-force algorithm.
3. Solve the system of market conditions as in the fixed crop set algorithm for new price guesses: this assumes that product sets do not respond to changing prices.
4. Solve individual farms' problems as in the brute-force algorithm with price guesses obtained from the fixed crop set GE system.
5. Evaluate convergence in product sets. If all farms' optimal product sets obtained in step 4 are the same as in step 2, stop. If not, return to step 2.

This algorithm exploits the low-dimensionality and low computational cost of solving for general equilibrium when product sets are fixed. By sequentially updating the farms' product sets, this algorithm converges to the same answer as the brute-force approach in just a few iterations.

D FIXED INPUT SHARES MODEL

Pipeline Π_3 in Section 6.3 freezes each input's allocation across a farm's existing crops at its solved current Π_2 shares. All crops therefore use the pooled technology, with land elasticity $\bar{\gamma}$ and flexible-input elasticities $\bar{\alpha}_g$. For each flexible input, define $s_{f,g,i}^0 = x_{f,g,i}^0 / \sum_{j \in I_f} x_{f,g,j}^0$, with analogous land shares $s_{f,l,i}^0$. The superscript 0 denotes the current pooled-technology allocation, which reproduces the observed farm-crop inputs and outputs. Shares and crop sets I_f remain fixed in the counterfactual, while farm input totals and the allocation across farms can change.

Let $X_{f,g}$ denote the farm's total flexible input g , and distinguish physical land demand L_f^d from the land endowment L_f^s appearing in its distorted land constraint. The farmer solves

$$\max_{\{y_{f,i}\}_i, L_f^d, \{X_{f,g}\}_g} \sum_{i \in I_f} (p_i y_{f,i})^\eta - \sum_{g=1}^G r_g \tau_{f,g} \sum_{i \in I_f} \tau_{f,g,i} s_{f,g,i}^0 X_{f,g} \quad (\text{D.1})$$

s.t.

$$y_{f,i} = z_{f,i} \left(s_{f,l,i}^0 L_f^d \right)^{\bar{\gamma}} \prod_g \left[\left(s_{f,g,i}^0 X_{f,g} \right)^{\bar{\alpha}_g} \right] \quad (\text{D.2})$$

$$\sum_{i \in I_f} s_{f,l,i}^0 \tau_{f,l,i} L_f^d = L_f^s \quad (\lambda_f). \quad (\text{D.3})$$

Productivity and distortions are those extracted under pooled technology in Π_2 , with distortions then changed by the policy rule. Freezing the current optimal shares preserves the current allocation. Counterfactual crop prices and factor rents clear the same CES goods markets and aggregate factor markets as in the flexible-share economy. The difference in gains between Π_3 and Π_2 measures the remaining within-farm adjustment margin after technology heterogeneity has been removed.

E DEMAND AND PRODUCER AGGREGATION

COMMON-GOOD DEMAND: Π_4 . Model Π_4 keeps the technology, crop-specific productivity and distortions, frozen shares, and crop-level objective of Π_3 . It replaces CES demand with one homogeneous good: every crop sells at the common numeraire price $p_i = 1$, and aggregate consumption equals additive output. Equilibrium requires

$$C = Y = \sum_f \sum_{i \in I_f} y_{f,i}, \quad \sum_f X_{f,g} = X_g^{agg} \quad \forall g, \quad \sum_f L_f^d = L^{agg}. \quad (\text{E.1})$$

Factor rents and the common component of land shadow costs adjust to clear these markets under the same distortion policy. No separate crop-demand equations remain. This is an exact common-good equilibrium, rather than an approximation using a large CES elasticity. Observed crop prices are normalized to one in the current economy, so replacing demand preserves current farm-crop quantities and the common observed output Y_0 .

FARM-LEVEL PRODUCTION: Π_5 . The final step sums observed crop outputs and inputs within each farm-season, applies the estimated pooled production function to the resulting single good, and re-extracts farm-level productivity and distortions. In particular, current single-product productivity is

$$z_f^S = \frac{\sum_{i \in I_f} y_{f,i}^0}{\left(\sum_{i \in I_f} l_{f,i}^0 \right)^{\bar{\gamma}} \prod_g \left(\sum_{i \in I_f} x_{f,g,i}^0 \right)^{\bar{\alpha}_g}}. \quad (\text{E.2})$$

The single-product model then applies the policy to its own extracted farm-level objects. Its producer objective applies η to aggregate farm revenue, replacing $\sum_i (p_i y_{f,i})^\eta$

with $(\sum_i p_i y_{f,i})^\eta$.

The placement of concavity can be seen directly under pooled technology and frozen shares. Write $y_{f,i} = a_{f,i} Q_f$, where

$$a_{f,i} = z_{f,i} (s_{f,l,i}^0)^{\bar{\gamma}} \prod_g (s_{f,g,i}^0)^{\bar{\alpha}_g}, \quad Q_f = (L_f^d)^{\bar{\gamma}} \prod_g X_{f,g}^{\bar{\alpha}_g}. \quad (\text{E.3})$$

At the common price, the ratio of the crop-resolved revenue term to the farm-level revenue term is

$$H_f = \frac{\sum_{i \in I_f} a_{f,i}^\eta}{\left(\sum_{i \in I_f} a_{f,i} \right)^\eta} \geq 1, \quad 0 < \eta \leq 1. \quad (\text{E.4})$$

The curvature difference vanishes when $\eta = 1$ or the farm grows one crop. Producer aggregation also remaps crop-level productivity and retained within-farm wedges into farm-level objects, so its contribution is broader than this curvature difference. The sign depends on how these changes affect expanding and contracting farms. Demand and producer aggregation are evaluated in that order; their separate contributions are conditional on this sequence.